

# Attack–Fault Identifiability and the Cost of Ambiguity at the Electronic Trading Gateway

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## Abstract

Operators of electronic trading gateways must often decide how to respond to suspicious order flow before knowing whether its cause is a cyber compromise, a software fault, or an ordinary market shock. We derive, for a linear-Gaussian model of gateway-observable order flow, an exact criterion — equality of severity-weighted projected offsets — under which a compromise and a fault are observationally indistinguishable, together with a matching cost floor: under this condition, no policy using the observation alone can beat the best constant action, imposing a strictly positive ambiguity cost relative to an oracle. We show this confounding is itself fragile under state-level (rather than output-level) dynamics, requiring the specific pair difference to lie in the classical unobservable subspace of the system — a condition that is trivial for a generic dynamics matrix, though we do not claim real, structured market dynamics are necessarily typical of that generic set. We then report a controlled empirical study, on the ABIDES agent-based market simulator, of how incident-construction, feature access, and calibration choices affect classification and decision cost — explicitly not a test of the theorem above, since none of the tested nonlinear constructions were confirmed to satisfy its exact premise. Across five profiles (four injection designs sharing one simulation infrastructure, one feature-engineering variant), classification and cost outcomes vary substantially with these choices; a full 50-cell cost audit, using direct class-conditional cost weighting, finds that the fitted classifier has a lower point-estimate cost than a naive always-contain policy in all 50 configurations (a supported improvement in 49) and is never worse than

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the best constant-action policy with a supported interval; most cells where it adds no supported value are in one profile (Study 5). We report this as a retrospective, whole-episode diagnostic study with a hypothetical decision-cost evaluation attached, establishing sensitivity to specific, named implementation choices — not a general claim about linear-to-nonlinear transfer difficulty, not a demonstrated real-time containment benefit (the classifier uses complete-episode features, so no claim is made about value delivered before an incident’s consequences finish accumulating), and not a claim to measured economic harm, since the underlying cost figures price a market-impact model against measured order volume, not independently validated incremental loss. We also show that applying a deployment-calibrated decision threshold to a classifier score trained at a different class balance (prior shift) can manufacture a spurious cost finding. The headline classification cells and the cost table are generated on paired seeds under a hardened, fingerprint-verified pipeline; the other simulator results predate it and are reported as exploratory. A practitioner checklist and full per-cell evidence tables are provided.

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*Keywords:* cyber-physical security, electronic trading, attack-fault identifiability, cost of ambiguity, decision under uncertainty, incident response, agent-based market simulation

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## 1. Introduction

A trading participant can generate harmful or suspicious orders because of a compromised credential, a software defect, its own order-generation policy, ordinary market conditions, or some combination of these. An operator — a broker or trading firm’s incident-response controller — must often act before the cause is known: throttle, cancel orders, revoke access, or isolate a process. Each of these actions has a cost if applied to a case that did not need it, and each carries risk if withheld from a case that did. **A scope note: this motivating scenario describes a real-time decision problem, but the theoretical and empirical results in this paper concern retrospective diagnosis from complete-episode observations, with a hypothetical decision-cost evaluation attached to the resulting classifications.** We use the real-time motivation to explain *why* the underlying

identifiability question matters operationally, not as a claim that this paper’s specific classifiers or cost figures measure the value of a timely intervention — Section 3.5 states the gap between these two framings precisely, and it should be read as qualifying every empirical claim in Sections 5–7, not as an isolated limitation confined to one paragraph.

Two questions precede any policy question. First: **can the causes even be told apart** from the evidence an operator is ordinarily permitted to observe (order acknowledgments, market data, session and process telemetry)? Second, if not: **how costly is that ambiguity**, given that operators must still act on it, typically via a fixed, threshold-based response? A fixed, threshold-based response answers the second question without the first: it trips a threshold (order rate, cancel ratio, loss limit) and acts, with no attempt to establish cause before acting.

This paper answers the first question with a formal identifiability result (Sections 3–4), scoped deliberately to a linear-Gaussian order-flow model — mirroring the approach in the closest structural analog to our problem, the literature on distinguishing cyberattacks from physical faults in cyber-physical systems, which likewise proves distinguishability results only for linearized or additive-unknown-input models. **Section 4.1 then shows this confounding is fragile even within the linear world itself**, once a more realistic state-level injection semantics is used, via an exact connection to classical control-theoretic observability. Sections 5–6 validate the core theorem at pilot scale within that same linear world. **Section 7 is the paper’s central empirical contribution**: a controlled study of how incident-construction choices (which observable statistics are matched between two mechanisms, which features an observer has access to, and how a fitted classifier’s decision threshold is calibrated) affect classification and decision cost in a realistic agent-based market simulator. This is explicitly not a test of whether the theorem above holds in a nonlinear market — the constructions studied were never shown to satisfy the theorem’s premise (equality of the full observation-trajectory law), so no result in Section 7 confirms, contradicts, or measures a “failure to transfer” of Theorem 1. What Section 7 does establish is that matching incident mechanisms on a small number of chosen summary statistics, in a realistic multi-agent market, leaves a highly construction-dependent residual of exploitable information in other channels — a finding about observation-matching practice, not about the linear theorem’s validity.

**Contributions.** 1. An exact fixed-instance equality criterion for observational indistinguishability between two incident mechanisms in a linear-

Gaussian order-flow model ( $\theta_K C b_K = \theta_F C b_F$ ), together with a separate rank-based characterization of which severity/direction pairs can be made to satisfy it (Theorem 1 and Corollary 1), together with a cost-of-ambiguity lower bound (Proposition 1). 2. A pilot-scale simulation consistent with both predictions within the linear model. 3. **A second, exact theoretical result (Theorem 2, Section 4.1) showing Theorem 1’s confounding is itself fragile even without leaving the linear model:** once the incident is modeled as a state-level rather than output-level disturbance — arguably the more realistic choice — confounding requires the far more restrictive condition that the specific pair difference  $\Delta_0$  lie in the classical (Kalman) unobservable subspace of the dynamics [1, 2], which is trivial for a generic  $(A, C)$  (though “generic” is a statement about a full-measure set of matrices, not a claim that real, structured market dynamics are typical of that set — Section 4.1 states this qualification explicitly), confirmed analytically and by a simulation in which an exactly-confounded pair becomes classifiable from the post-onset part of each episode under generic coupled dynamics, while it stays at chance under output-level injection. 4. **A controlled empirical study on ABIDES of how incident-construction choices affect classification and decision cost**, escalated into a systematic sensitivity analysis across four injection designs (parameterizations of one shared infrastructure): the accuracy ranking between a summary-matched and a contrast construction is itself highly sensitive to which features an observer has access to, which specific statistic was matched during calibration, classifier capacity, and proper threshold calibration under prior shift (Section 6) — with five qualitatively distinct classification profiles observed across four designs. **This is a study of observation-matching and decision-cost behavior, not a test of Theorem 1:** none of the tested constructions were shown to instantiate the theorem’s premise (equality of the full observation-trajectory law), so no claim here concerns the theorem’s validity, only the practical difficulty of engineering a nonlinear analog of exact confounding and the decision-cost consequences of not fully succeeding at it. 5. **A duration/persistence ablation (Section 7.8)**, which does not causally isolate the mechanism it was designed to test: removing incident persistence lowered classification accuracy in an exploratory run, directionally consistent with Theorem 2’s state-feedback prediction, but the design conflates persistence with incident duration and cumulative activity, and a follow-up isolation redesign (Section 7.10.2) remains inconclusive at the sample size tested. 6. A narrower statement of what results of this form (ours and the cyber-physical-systems analog

it is built on) can and cannot be assumed to say about real deployed systems, and a question left to the companion paper (Paper 2) about the cells where the fitted classifier adds no supported value over the best constant policy, at the low end of the prior grid or in the Study 5 profile (Section 7.7); Section 7.11 states that question and what this paper does not claim about it.

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## 2. Related work

**Decision-theoretic cyber response.** Belief-state (POMDP) formulations of cyber defense track an attacker’s progression through a network and choose interference actions under noisy alerts (Miehling, Rasouli & Teneketzis [3]); related work formulates response as a Stackelberg game over attack-response trees (Zonouz et al.’s RRE [4]) or balances service continuity against intrusion risk under partial observability (Hammar & Stadler [5]; see also their learning-based extensions against dynamic attackers and under model misspecification [6, 7], and the workshop paper of Booker & Musman [8]). None of this literature prices market liquidity or service interruption to legitimate trading. Our own cost structure is a fixed-loss-table/market-impact-proxy analysis (Sections 3.4, 6, 6.3), not a quantified model of endogenous liquidity replacement by other market participants, so the claim is narrower: this literature does not price market-impact costs of the kind Section 6 constructs, not that it omits a component this paper has itself quantified.

**Cyber-physical fault-versus-attack differentiation.** The closest structural analog to our identifiability question is the literature distinguishing physical faults from cyberattacks in power systems and other cyber-physical systems. Pasqualetti, Dörfler & Bullo [9] is a foundational precedent here, arguably closer to our Theorem 2 than any other cited work: they characterise, with geometric control theory, which attacks are undetectable or unidentifiable by an output-based monitor (those that excite only the system’s zero dynamics); Theorem 2 uses the attack-free special case of this object, the Kalman unobservable subspace, to characterize when compromise-versus-fault confounding survives state feedback. Roy & Dey [10] is a more direct methodological precedent for our Theorem 1 specifically: they model an anomalous input to a linear system, estimate it via a sliding-mode unknown-input observer, and derive a closed-form distinguishability metric with a criterion for when the estimated input is provably a fault versus an attack,

including a “fault-mimicking cyberattack” corollary — an algebraic condition under which an attack direction lies in the fault subspace, the direct analog of our Corollary 1. Observer-based schemes that detect and isolate faults and deception attacks simultaneously [11], and detectors designed to keep faults and attacks distinguishable in interconnected systems [12], address the design side of the same question; neither prices the response or treats a trading setting. On the data-driven side, Anwar, Mahmood & Shah [13] separate simulated faults from attacks on a standard IEEE bus test system with a PCA projection and a support-vector classifier, and later systems fuse network and process data for attack-versus-failure attribution [14]. Closest in spirit to our decision-theoretic framing, a recent triage framework for energy cyber-physical systems attributes detected anomalies to physical faults or to measurement-integrity loss from physical-consistency features, prices misattribution with explicit response-cost matrices against blanket responses, and observes that exactly matched fault/attack pairs are discriminated at chance level [15]. That work is empirical and set in process and wind-farm control; it does not derive a necessary-and-sufficient indistinguishability condition, a cost floor under confounding, or the state-feedback fragility of confounding, and it does not address electronic trading. Its matched-pair observation is the empirical counterpart of our Theorem 1 and Proposition 1, whereas our Section 7 concerns the harder case in which pairs are matched only on summary statistics. Proposition 1 provides a mechanism by which classifiers *can* fail when the confounding condition holds; we have not checked any published dataset against that condition.

**Engaging this literature’s methodology directly, not just its conclusions:** Roy & Dey’s validation, like our own linear-model pilot (Section 5), is simulation-based rather than tested against a physically realized, nonlinear testbed — their “simulation case studies” instantiate the same linear model their theorem is proved for, so their framework’s internal consistency is well-established, but whether an analogous criterion could be validly instantiated in a genuinely nonlinear physical system (analogous to our ABIDES study) is, to our knowledge, not attempted in their work or elsewhere in this literature. Our Section 7 takes up this open question empirically and reports that constructing a valid nonlinear analog of exact confounding — one that actually matches the full observation law, not merely a chosen summary statistic — is itself a nontrivial, easy-to-get-wrong engineering task, with the resulting classification ranking sensitive to feature access, calibration target, and classifier capacity in ways the linear model does not anticipate. We do

not claim to have established whether their criterion’s qualitative implications hold or fail once genuinely instantiated in a nonlinear system, since none of our attempted constructions were confirmed to be valid instantiations; what we do show is that attempting the instantiation is harder, and more sensitive to implementation choices, than the linear theory alone would suggest. We read our contribution, in this literature’s terms, as raising and empirically probing — without fully resolving — a validation question their own framework does not itself address: what does it take to correctly realize a linear-derived distinguishability criterion’s premise once observations come from genuinely nonlinear, compounding dynamics rather than being assumed linear from the outset?

Stealth that holds in a linear model need not hold in a nonlinear one: stealthy injections must be redesigned for nonlinear differential-algebraic grid models [16]. A separate, more recent line (Alisic & Amin, a preprint [17]) approaches a related transfer question geometrically rather than empirically via simulation: they extend sensor-attack analysis from linear state spaces to dynamics on Lie groups and show that an attack transfers across operating conditions, keeping its impact and stealth, only if it commutes with the flow (a Lie-bracket condition); for instance, a sensor attack that is stealthy under one motion (e.g., straight-line driving) can become detectable under another (e.g., a turn), because non-commuting motion rotates the attack relative to the state, so that transferability is a geometric constraint to be satisfied jointly with stealthiness. Their result and ours are related but not directly comparable: theirs is a formal analysis of a specific transfer property in a specific technical setting; ours is an empirical study of how difficult it is to engineer a nonlinear analog of exact confounding in a realistic market, without deriving or checking an analogous condition for our setting. **We do not claim that our market satisfies or fails their condition, or any analogous one** — no such condition has been derived for the order-flow/observation-matrix structure of Section 3, so no claim about satisfying or failing it is supported. The genuine open question, stated at the correct strength, is whether a Lie-group-style transfer condition *could* be formulated for our setting, and if so, whether it would help explain the construction-dependence Section 7 documents — a question this paper raises but has not answered.

**Market manipulation detection and trading kill switches.** A large literature detects manipulative patterns (spoofing, wash trading) from trade and order-book data (recent supervised and unsupervised deep-learning de-

tectors are typical [18]); this literature classifies trading patterns as manipulative or not and does not ask whether a compromised gateway and a faulty one can be told apart, nor price the response. **A threshold-based anomaly trigger (for instance on an order-flow statistic or a cancel ratio) is an *observation-dependent* policy, responding to data, not the *observation-independent constant* policy (“always contain”) this paper uses as one baseline.** The existence of threshold triggers does not, by itself, validate a constant action as a realistic operational comparison — the two are related (both act without first diagnosing cause) but are not the same object, and a reader wanting an operational comparison would need a specified threshold detector with comparable feature access and timing, which this paper does not provide. The constant-action baselines in this paper should be read as decision-theoretic reference points (the best achievable without any observation at all), not as a claim that real kill-switches are literally constant policies. Separately, empirical work on automating stock-market manipulation via compromised brokerage accounts [19] shows that manipulative order generation can be automated through compromised brokerage accounts, supporting our model’s treatment of compromise as a distinct, structured perturbation to order flow rather than an unstructured anomaly.

**Costly information acquisition.** A separate literature treats costly, endogenous information acquisition in asset markets (e.g., models in the Grossman–Stiglitz tradition [20], and recent optimal-stopping formulations of costly sequential information acquisition in general decision problems [21]). This addresses *traders’* incentives to learn about asset value, not an *operator’s* decision to verify an incident’s cause before containing it, and does not engage with observational indistinguishability at all; we treat it as a technique-adjacent rather than a competing literature.

**Scope of the novelty claim, stated precisely rather than sweepingly.** We do not claim no existing work touches any combination of identifiability analysis, market cost, and fixed-policy critique — the CPS distinguishability literature (Roy & Dey; Anwar, Mahmood & Shah) and the recent cost-aware triage work in energy CPS [15] and the Lie-group transfer literature (Alisic & Amin) share substantial theoretical machinery with Sections 3–4, and the market-manipulation-detection literature addresses an adjacent part of our motivating setting. What we have not found is the specific combination this paper assembles: a formally exact identifiability and cost-floor result for the compromise-versus-fault question *in the electronic-*

*trading setting specifically*, together with a controlled empirical study of how incident-construction choices affect classification and decision cost in a realistic nonlinear market simulator, and a decision-cost benchmark distinguishing the best-achievable-using-observations risk from the best-constant-action risk. This is narrower than an unqualified claim that no existing work combines these elements.

**Incremental contribution relative to the two closest predecessors.** Roy & Dey [10] derive a fault-mimicking distinguishability criterion via a sliding-mode unknown-input observer and validate it in linear simulation; Pasqualetti, Dörfler & Bullo [9] establish the geometric (zero-dynamics) machinery for attack detectability and identifiability whose attack-free special case, the unobservable subspace, Theorem 2 uses. Neither of these — nor, to our knowledge, any other work in this literature — does the following, which is what this paper actually adds: (1) applies this style of criterion to the specific loss structure of a trading gateway (compromise vs. fault vs. an oracle, with a market-impact-based cost floor, rather than a generic detection-accuracy or false-alarm-rate metric); (2) uses the attack-free special case (the unobservable subspace) of their machinery to compare two injection semantics for the same incident pair: a pair that is exactly confounded under output-level injection is, for generic dynamics, not confounded under state-level injection (Theorem 2), so Theorem 1’s output-level criterion is not a stable target once the incident propagates through the dynamics; (3) examines empirically, rather than assuming, whether summary-matching produces the confounding the linear criterion describes in a nonlinear multi-agent simulator, across four injection designs that are parameterizations of one shared infrastructure (Section 9). **Items (1) and (2) are careful applications of established machinery rather than new theory; (3) is the paper’s main empirical contribution.** The prior-shift threshold correction used in Section 6 is standard (Elkan [22]; Sacerens et al. [23]); we report its effect on this pipeline’s error rates as a finding, not as a methodological contribution.

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### 3. Model

**Domain grounding for the  $K/F$  labels:** these labels name programmed order-generation mechanisms in the formal model and the ABIDES simulation, not a general claim about compromise

**or fault in the wild.** Throughout this paper,  $K$  (“compromise”) and  $F$  (“fault”) denote specific, fully-specified perturbations to the order-flow process (Sections 3.2 and 7’s ABIDES designs) — a directional bias, an elevated order rate, or an altered cancellation pattern, applied under a chosen severity and onset time. We do not claim these mechanisms are representative of any actual attacker’s access, objective, or capability, nor of any specific real software defect; an attacker could in principle imitate a fault’s statistical signature, and a fault could in principle produce strongly directional order flow, either of which would blur the distinction this paper’s labels assume is clean. What this paper’s identifiability and classification results establish is *behavioral-mechanism identification* — whether two specified generative processes are distinguishable from gateway-observable order flow — not *inference about intent or authorization*, which would require a different kind of evidence entirely (e.g., credential logs, code-change records) that this paper does not model. The four ABIDES designs used in Section 7 are best understood as four independently-*parameterized* scenario variants sharing common simulation infrastructure, not four independent software implementations, nor a representative sample of the space of possible real incidents (Section 9’s independence-classification item states this explicitly).

### 3.1. Order-flow dynamics

Let  $x_t \in \mathbb{R}^n$  denote a latent representation of normal order-flow state (e.g., components loading onto price pressure, volume intensity, and order-cancellation propensity), evolving under known linear-Gaussian dynamics with transition matrix  $A$  and process-noise covariance  $Q$ . The complete model, including the incident injection and the gateway observation described next, is stated in Definition D1 below.

### 3.2. Incident injection

At an onset time  $\tau$ , possibly unknown to the operator, an incident of mechanism  $M \in \{K, F\}$  (compromise or fault) injects an additive perturbation into the pre-observation order-flow signal. The injection direction  $b_M$  describes how a compromise versus a fault characteristically distorts order flow; the scalar  $\theta_M$  is its severity or magnitude.

**Definition D3 (Mechanism instance).** A mechanism instance is a triple  $(M, \theta_M, b_M)$  with cause  $M \in \{K, F\}$  ( $K$ : compromise,  $F$ : fault), a deterministic severity  $\theta_M \in \mathbb{R}$  and a deterministic injection direction  $b_M \in \mathbb{R}^n$ . Statements compare one fixed instance of  $K$  with one fixed instance of

$F$  (pairwise-instance reading); no distribution over severities, directions or onsets within a cause is modelled.

### 3.3. Gateway-observable output

The operator observes only a lower-dimensional ( $m < n$ ), noisy projection of order flow through a fixed, known observation matrix  $C \in \mathbb{R}^{m \times n}$ , representing what a gateway or market-data feed can actually surface (e.g., spread, depth, order-cancellation rate). In the model of this section the incident shifts the signal at the point of observation only; Section 4.1 considers the alternative in which the perturbation propagates through the dynamics.

**Definition D1 (Output-level injection model).** Fix  $n, m$ , a matrix  $A \in \mathbb{R}^{n \times n}$ , an observation matrix  $C \in \mathbb{R}^{m \times n}$ , covariance matrices  $Q \in \mathbb{R}^{n \times n}$  and  $\Sigma \in \mathbb{R}^{m \times m}$ , and an onset time  $\tau$ . Under a mechanism instance  $(M, \theta_M, b_M)$  (Definition D3), the latent state, the perturbed state and the observation evolve for  $t \geq 0$  as

$$\begin{aligned} x_{t+1} &= Ax_t + w_t, & \tilde{x}_t &= x_t + \mathbb{1}[t \geq \tau] \theta_M b_M, \\ y_t &= C\tilde{x}_t + \varepsilon_t, \end{aligned}$$

with  $w_t \sim \mathcal{N}(0, Q)$  and  $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$ . The injection enters the observation only; the state recursion does not see it. (Section 3 additionally takes  $m < n$ ; no statement below uses that restriction.)

Everything except the injection term is common across hypotheses about  $M$ . The results of Section 4 rest on the following standing assumptions, each of which the implementation satisfies. Independence of the noise sequences, (A1), is a sufficient, and stronger, way of obtaining the jointly Gaussian observation trajectory than joint Gaussianity alone. The initial state is implemented as the deterministic point  $x_0 = 0$ , (A2). The onset need not be known to the operator, (A4). The finite-horizon experiments in Sections 5–6 always include timesteps at and after the incident onset  $\tau$  by construction ( $T = 40$ ,  $\tau = 20$ , so  $T > \tau$  in every run), so (A5) holds; on a window ending before  $\tau$ , Theorem 1’s distinguishability claim would be vacuous.

#### Standing assumptions.

**(A1) Gaussian, independent noise.**  $w_t \sim \mathcal{N}(0, Q)$  and  $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$ , with  $Q$  and  $\Sigma$  symmetric positive semidefinite; the family  $\{w_t\}_{t \geq 0} \cup \{\varepsilon_t\}_{t \geq 0}$  is mutually independent (across time and across the two sequences), and its law does not depend on the instance.

**(A2) Deterministic initial state.**  $x_0 = 0$ . (The proofs use only that  $x_0$  is Gaussian, possibly degenerate, independent of the noise, with a law that does not depend on the instance;  $x_0 = 0$  is the implemented special case.)

**(A3) Known, common model.**  $A, Q, C, \Sigma$  are known and identical under both causes; only the injection term  $\theta_M b_M$  depends on the instance.

**(A4) Common onset.** The onset  $\tau$  is deterministic, common to both causes, not necessarily known to the operator. (The proofs use only that  $\tau$  is deterministic and common; no statement requires a policy to know  $\tau$ .)

**(A5) Window reaches the onset.** The observation window  $W$  contains at least one  $t \geq \tau$ . Theorem 2 uses the stronger form  $W \supseteq \{t : t \geq \tau\}$  and Theorem 3 the stronger form  $W \supseteq \{\tau, \dots, \tau + H - 1\}$ .

**(A6) Positive-definite observation noise.**  $\Sigma \succ 0$ . Used only to apply Proposition 3 to the model on a finite window: it makes the covariance of the stacked observations positive definite ( $V \succ 0$ ). No other statement uses a positivity condition on  $\Sigma$  or  $Q$ ;  $Q$  is only required to be positive semidefinite.

Identifiability is a property of the laws of the observation trajectory, which the following definitions make precise.

**Definition D4 (Observation window; observational indistinguishability).** An observation window is a set  $W \subseteq \{0, 1, 2, \dots\}$  of time indices, finite or not; the observation trajectory on  $W$  is  $Y_W = (y_t)_{t \in W}$ . Write  $P_M^W$  for the law of  $Y_W$  under instance  $(M, \theta_M, b_M)$  (for infinite  $W$ , the law on the product  $\sigma$ -algebra, determined by its finite-dimensional marginals). Instances  $(K, \theta_K, b_K)$  and  $(F, \theta_F, b_F)$  are observationally indistinguishable on  $W$  if  $P_K^W = P_F^W$ . Theorem 1 uses  $W = \{t : t \geq 0\}$ ; “indistinguishable at every  $t \geq \tau$ ” (Theorem 2) means  $P_K^W = P_F^W$  for  $W = \{t : t \geq \tau\}$ ; Theorem 3 (the finite-horizon version of Theorem 2) uses  $W = \{\tau, \dots, \tau + H - 1\}$ .

**Definition D5 (Distinguishable in the population limit).** Two instances are distinguishable in the population limit on  $W$  if they are not observationally indistinguishable on  $W$ , i.e.  $P_K^W \neq P_F^W$  (equivalently  $\text{TV}(P_K^W, P_F^W) > 0$ , Definition D10). This is a statement about the two laws only; it asserts nothing about the accuracy attainable from finitely many observations.

**Definition D11 (Gaussian discriminability).** If  $Y \mid M \sim \mathcal{N}(\mu_M, V)$ ,  $M \in \{K, F\}$ , with a common covariance  $V \succ 0$ , let  $\delta = \mu_K - \mu_F$  and

$d = \sqrt{\delta^\top V^{-1} \delta} \geq 0$  (the whitened mean separation).

### 3.4. The loss table, and what it assumes

The cost analysis in Sections 6–7 uses the two-action, two-cause loss table

| Action      | Compromise $K$ | Fault $F$ |
|-------------|----------------|-----------|
| Contain     | $C_{OP}$       | DISRUPT   |
| Leave alone | HARM           | 0         |

The table is a special case of a general two-action, two-cause table; Proposition 1 keeps the symbols above. The decision problem built on the table, and the distance between observation laws used to extend it beyond exact confounding, are defined as follows.

**Definition D8 (Loss table).** The general two-action, two-cause loss table  $L(a, M)$ ,  $a \in \{\text{contain, leave alone}\}$ ,  $M \in \{K, F\}$ , is

$$\begin{aligned} L(\text{contain}, K) &= C_K, & L(\text{contain}, F) &= C_F, \\ L(\text{leave alone}, K) &= H_K, & L(\text{leave alone}, F) &= H_F, \end{aligned}$$

with  $C_K, C_F, H_K, H_F \in \mathbb{R}$ . This paper’s table (Section 3) is the special case

$$C_{OP} = C_K, \quad \text{DISRUPT} = C_F, \quad \text{HARM} = H_K, \quad H_F = 0.$$

Propositions 1 and 2 are stated in this paper’s symbols  $C_{OP}$ , DISRUPT, HARM.

**Definition D9 (Prior, policies, Bayes risk, oracle risk, cost of ambiguity).** The cause is drawn as  $M = K$  with prior probability  $p = P(K)$  and  $M = F$  with probability  $1 - p$ ; given  $M$ , the observation trajectory  $Y_W$  has law  $P_M^W$ . A policy is a measurable map  $\pi$  from observation trajectories on  $W$  to  $[0, 1]$ , read as the probability of the action *contain* (randomized policies allowed; a deterministic policy takes values in  $\{0, 1\}$ ). Its expected cost is

$$\begin{aligned} R(\pi) &= p \mathbb{E}_K [\pi(Y_W) C_K + (1 - \pi(Y_W)) H_K] \\ &\quad + (1 - p) \mathbb{E}_F [\pi(Y_W) C_F + (1 - \pi(Y_W)) H_F]. \end{aligned}$$

The best achievable expected cost is  $R^* = \inf_\pi R(\pi)$ ; the oracle, which observes  $M$ , has expected cost  $R_{\text{oracle}} = p \min\{C_K, H_K\} + (1 - p) \min\{C_F, H_F\}$ ; the cost of ambiguity is  $\text{CoA}^* = R^* - R_{\text{oracle}}$ . Where the observation is the raw trajectory  $Y$  the subscript  $Y$  is written ( $R_Y^*$ ,  $\text{CoA}_Y^*$ ).

**Definition D10 (Observation densities; total variation).** Let  $\mu$  be a  $\sigma$ -finite measure dominating  $P_K$  and  $P_F$  (e.g. the finite measure  $P_K + P_F$ ), and  $f_K, f_F$  their densities with respect to  $\mu$ ;  $\int \cdot dy$  denotes integration against  $\mu$ . The total variation distance is

$$\text{TV}(P, P') = \sup_B |P(B) - P'(B)| = \frac{1}{2} \int |f - f'| dy,$$

so that  $1 - \text{TV}(P_K, P_F) = \int \min\{f_K, f_F\} dy$ .

**This table encodes a specific scenario, not a general property of compromises and faults, and we state that scenario explicitly rather than leaving it implicit:** it assumes there exists a specific class of fault for which containment is unnecessary and leaving the system running incurs zero *incremental* loss relative to doing nothing — for example, a fault that produces anomalous-looking but ultimately harmless order-flow noise (a logging error inflating apparent order counts, or a benign retry loop that does not cross the spread), as opposed to a fault that itself generates destructive orders requiring intervention regardless of cause. The paper’s motivating discussion elsewhere also mentions harmful software defects that plausibly *would* require containment — for that broader class of fault, the correct loss table has DISRUPT replaced by whatever cost containing a harmful-but-benign-cause incident actually carries, and the fault’s own leave-alone cost would not generally be zero. **The strict value of distinguishing the two causes is zero whenever some single action is simultaneously optimal for both** — Proposition 1’s cost floor is a property of *this* loss structure (and, more generally, of any loss structure where the causes call for genuinely different optimal actions), not evidence that ambiguity about cause is always costly regardless of what the two causes actually require.

**Audited against the actual simulated fault mechanisms, this assumption largely does not hold for the ABIDES results in Section 7.** Every one of the four ABIDES designs’ fault ( $F$ ) mechanisms submits real limit orders unconditionally (`place_incident_order()` calls `place_limit_order()` for  $K$ ,  $F$ , and baseline alike, confirmed by direct source inspection) — there is no “harmless, logging-only” fault anywhere in this project’s actual implementation, only the illustrative logging-error/retry-loop example given above, which does not describe any simulated mechanism. Auditing each design’s specific  $F$  parameterization against whether it is non-directional, non-elevated in size/rate, and free of price aggressiveness (the properties that would make “leave alone” plausibly costless), per design: V1 confounded:

fails (directionally biased and volume/price-elevated); V2 confounded: fails (same reasons); V3 confounded: questionable (non-directional, but meaningfully elevated order volume); V4 confounded: questionable (same reasons as V3). **Only the contrast regime’s  $F$  mechanism in V2 and V3 specifically (non-directional, no size/rate/price elevation, only added cancellation churn) plausibly satisfies the zero-incremental-loss assumption as stated** — V1’s and V4’s contrast-regime  $F$  mechanisms are also volume-elevated and therefore also questionable, not clean. In short: 0 of 8 audited cells (4 designs  $\times$  2 regimes) are clean passes without qualification; 2 of 8 (V2 and V3’s contrast regimes) are plausible passes; the remaining 6 range from questionable to a clear failure of the assumption.

**What this means for Section 7’s dollar-cost tables:** the illustrative cost reconstruction in Sections 6.2 and 7.7 is defensible as a hypothetical decision-cost scenario — a chosen loss table applied consistently to observed classification outcomes — but should not be read as an economically accurate costing of these specific simulated incidents, since most of them do not satisfy the scenario’s own defining assumption (that leaving  $F$  alone is actually free), and since **the underlying quantity itself is a modeled execution-cost proxy applied to submitted order volume, not a directly measured economic loss** — submitted volume need not equal executed volume (cancellations and partial fills intervene), and positive execution costs are not the same claim as positive *incremental* harm caused by  $F$  specifically, since ordinary baseline trading also has positive execution costs. Readers should treat Section 7’s dollar figures as “what this specific illustrative loss table implies, applying a market-impact cost model to measured submitted volume,” not as validated economic costs of the actual simulated fault behavior or its incremental harm relative to a baseline. **Section 6.3 prices the corrected loss table: a costing from directly-measured submitted order volumes (not executed volume, and not independently validated as incremental harm) shows  $F$ ’s order flow is nonzero-cost under this model, contrary to the original loss table’s zero assumption, at a scale (from \$850 to \$104,739 per episode, depending on design and regime; Table 1) that is small beside the illustrative cost triple’s other terms for Study 5 and the third design but large enough in the original design’s summary-matched regime to lower the cost-optimal threshold  $q^*$  (Table 1).** This simplification does not leave the preferred policy unchanged in every configuration: in the full 50-cell audit (Section 7.7), the classifier does not beat the best constant policy at point

estimate in 6 cells (5 of them Study 5), and its advantage over the best constant policy is not a supported improvement in 10 cells (Table 4), though every cell has a lower point-estimate cost than always-contain. Those cells lie at the low end of the prior grid or in the Study 5 profile.

### 3.5. Observation timing and scope

The motivating framing in Section 1 emphasizes acting *before* a full diagnosis is possible — an urgent-response setting. **The formal model and pilot experiments in this paper do not fully match that framing, and we state the mismatch explicitly rather than leave it implicit.** Theorem 1 and Proposition 1 concern the full infinite (or fixed-horizon) observation trajectory  $\{y_t\}_{t \geq 0}$ ; Section 5’s pilot uses whole-episode summary statistics (mean, standard deviation, net drift over the full 40-step window) as features, and Section 7’s ABIDES experiments likewise use full-episode summaries rather than a prefix available at some specific, pre-committed decision deadline. **This is, as implemented, closer to an offline or retrospective diagnosis study than to a real-time response-deadline study:** nothing in the current pipeline enforces that features are computed only from information available at the moment an operator would actually have to act, and the classifier is not “charged” for including post-decision information the way a genuinely online policy would need to be. A paper making a strict real-time-response claim would need to define an explicit observation cutoff and decision deadline, and evaluate classifiers using only prefix features available at that cutoff — we have not done this, and the results in Sections 5–7 should be read as characterizing what is discoverable given the full episode, not what is discoverable in time to inform the specific urgent action Section 1 motivates. This does not invalidate the identifiability results (which are about what a trajectory’s full distribution reveals, a well-posed question independent of timing), but it does mean the cost figures in Sections 6–7 should not be read as measuring the value of information available at a real operational decision point without this additional restriction.

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## 4. Theoretical results

**Theorem 1 (Indistinguishability criterion).** Under the model of Section 3, with common onset time  $\tau$ , mechanism instances  $(K, \theta_K, b_K)$

and  $(F, \theta_F, b_F)$  induce identical distributions over the observation trajectory  $\{y_t\}_{t \geq 0}$  if and only if

$$\theta_K C b_K = \theta_F C b_F. \quad (*)$$

*Theorem 1 uses* Definition D1, Definition D3, Definition D4 (window  $W = \{t : t \geq 0\}$ ) and assumptions (A1)–(A5).

*Proof.* Work in the output-level injection model (Definition D1) with window  $W = \{t : t \geq 0\}$  (Definition D4); by (A4) the onset  $\tau$  is deterministic and common to both instances (Definition D3), so (A5) holds for this  $W$ .

*Step 1 (decomposition).* Under Definition D1 the state recursion does not see the injection, so  $x_t = A^t x_0 + \sum_{s=0}^{t-1} A^{t-1-s} w_s$  under either instance. Hence, for every  $t \geq 0$ ,

$$y_t = G_t + m_M(t), \quad G_t = C x_t + \varepsilon_t, \quad m_M(t) = \mathbb{1}[t \geq \tau] \theta_M C b_M.$$

The process  $G = (G_t)_{t \geq 0}$  is a fixed measurable function of  $(x_0, \{w_t\}, \{\varepsilon_t\})$  through  $A$  and  $C$ . By (A1) and (A2) the law of  $(x_0, \{w_t\}, \{\varepsilon_t\})$  does not depend on the instance, and by (A3) neither do  $A$  and  $C$ ; therefore the law of  $G$  is the same under both instances. The sequence  $m_M$  is deterministic (Definition D3, (A4)).

*Step 2 (if).* Suppose  $\theta_K C b_K = \theta_F C b_F$ . Then  $m_K = m_F =: m$ , and under either instance  $Y_W = G + m$ . For every finite  $S \subseteq W$ , the law of  $(y_t)_{t \in S}$  is the image of the (common) law of  $(G_t)_{t \in S}$  under the translation by  $(m(t))_{t \in S}$ , hence the same under both instances. Laws on the product  $\sigma$ -algebra that agree on all finite-dimensional marginals are equal (Definition D4), so  $P_K^W = P_F^W$ .

*Step 3 (only if).* Suppose  $P_K^W = P_F^W$ . Each  $G_t$  is Gaussian by (A1) and (A2), so  $y_t$  is integrable and equal laws give equal means:  $\mathbb{E}_K[y_t] = \mathbb{E}_F[y_t]$  for all  $t \in W$ . Since  $\mathbb{E}[G_t]$  does not depend on the instance (Step 1),  $m_K(t) = m_F(t)$  for all  $t \in W$ . Taking  $t = \tau \in W$  gives  $\theta_K C b_K = \theta_F C b_F$ , i.e. (\*).

*Remarks.* (i) The covariance of  $Y_W$  is that of  $G$ , which does not depend on the instance; this is the only role of the noise. (ii) Gaussianity is used only for integrability in Step 3; Step 2 uses only that the law of  $G$  is instance-independent. (iii) If  $W$  contained no  $t \geq \tau$ , that is, if (A5) failed, both laws would always coincide and the “only if” direction would fail; this is why (A5) is listed.  $\square$

**Corollary 1 (Fault-mimicking compromise — corrected for degenerate cases).** Fix  $u = Cb_K$ ,  $v = Cb_F$ , and a nonzero compromise severity  $\theta_K \neq 0$ . If the fault severity  $\theta_F$  may be any real number, then

$$\exists \theta_F \in \mathbb{R} : \theta_K u = \theta_F v \iff u \in \text{span}(v).$$

*Corollary 1 uses* Definition D3 and condition (\*) of Theorem 1; the equivalence itself is linear algebra and uses no probabilistic assumption.

*Proof.* Let  $u = Cb_K$ ,  $v = Cb_F$  and  $\theta_K \neq 0$ , and recall  $\text{span}(v) = \{\lambda v : \lambda \in \mathbb{R}\}$  (which is  $\{0\}$  if  $v = 0$ ).

( $\Rightarrow$ ) If  $\theta_K u = \theta_F v$  for some  $\theta_F \in \mathbb{R}$ , then, since  $\theta_K \neq 0$ ,  $u = (\theta_F/\theta_K) v \in \text{span}(v)$ .

( $\Leftarrow$ ) If  $u \in \text{span}(v)$ , write  $u = \lambda v$  with  $\lambda \in \mathbb{R}$  and set  $\theta_F = \theta_K \lambda$ . Then  $\theta_F v = \theta_K \lambda v = \theta_K u$ .

The argument is linear algebra only (Definition D3 fixes that severities are real and directions are vectors). Combined with Theorem 1, the  $\theta_F$  so obtained makes the instances  $(K, \theta_K, b_K)$  and  $(F, \theta_F, b_F)$  observationally indistinguishable under the hypotheses of Theorem 1.  $\square$

Two points of precision: (i)  $\theta_K = \theta_F = 0$  trivially satisfies (\*) for *any*  $b_K, b_F$  and is excluded by fixing  $\theta_K \neq 0$  as above; (ii) linear dependence corresponds to  $\text{rank} \leq 1$ , not  $\text{rank} = 1$  exactly — both  $u$  and  $v$  could be zero, or one could be zero while the other is not, and these subcases behave differently (if  $v = 0$  but  $u \neq 0$ , no  $\theta_F$  satisfies the equation; if both are zero, every  $\theta_F$  trivially does). If both severities are additionally required to be nonzero, the condition sharpens to: either  $u = v = 0$ , or  $u, v$  are both nonzero and collinear. If severities must be positive, nonzero  $u, v$  must lie on the same ray, not merely the same line. For a *family* of admissible fault directions  $\{b_F^{(i)}\}$ , we distinguish two claims: **choosing** one fault direction gives fault-mimicking whenever  $u$  lies on the line through some single  $Cb_F^{(i)}$  (a union of lines); **superposing** multiple fault directions (if the generative model allows combined faults) gives the weaker requirement  $u \in \text{span}\{Cb_F^{(i)}\}$  (a full subspace or cone). The manuscript's claims use the span form and should be read as assuming fault superposition is a modeled possibility, not merely a choice among single fault directions.

We also flag, rather than resolve, a scope question: establishing that *some* instance  $(F, \theta_F, b_F)$  matches *some* instance  $(K, \theta_K, b_K)$  is a statement about overlapping parameter families, not a statement that  $P(Y \mid M = K) =$

$P(Y \mid M = F)$  after marginalizing over a distribution on severities, onset times, or directions within each mechanism class. Different fixed severities on the same projected direction produce different means; a composite/marginal indistinguishability claim requires either a pairwise-instance reading (which is what Theorem 1 and this corollary literally give) or an explicit mixture model over the mechanism class, which this paper does not specify. All results below should be read as pairwise-instance statements unless a mixture model is given.

**Corollary 2 (Separation implies distinguishability, not high accuracy).** If  $\text{rank}([Cb_K \ Cb_F]) = 2$ , no pair  $(\theta_K, \theta_F)$  other than  $(0, 0)$  satisfies  $(*)$ ; for any instance pair with  $(\theta_K, \theta_F) \neq (0, 0)$ ,  $K$  and  $F$  are then distinguishable in the population limit.

*Corollary 2 uses* Definition D3, Definition D4, Definition D5 and, through Theorem 1, Definition D1 and assumptions (A1)–(A5).

*Proof.* Let  $u = Cb_K$ ,  $v = Cb_F$  and suppose  $\text{rank}[u \ v] = 2$ , i.e.  $u$  and  $v$  are linearly independent (this requires  $m \geq 2$ ). If  $\theta_K u = \theta_F v$ , then  $\theta_K u - \theta_F v = 0$  and linear independence gives  $\theta_K = \theta_F = 0$ . Conversely  $(0, 0)$  satisfies  $(*)$  trivially. Hence no pair  $(\theta_K, \theta_F)$  other than  $(0, 0)$  satisfies  $(*)$ , which is the first clause. (This step is linear algebra only and uses no probabilistic assumption.)

Now take an instance pair (Definition D3) with  $(\theta_K, \theta_F) \neq (0, 0)$ . By the previous paragraph  $(*)$  fails, so by Theorem 1 (under Definition D1 and (A1)–(A5), window  $W = \{t : t \geq 0\}$ , Definition D4)  $P_K^W \neq P_F^W$ : the instances are distinguishable in the population limit (Definition D5).

The hypothesis  $(\theta_K, \theta_F) \neq (0, 0)$  cannot be dropped: for  $\theta_K = \theta_F = 0$  both instances have the uninjected law, whatever  $b_K, b_F$ . The conclusion concerns the two laws only (Definition D5) and gives no bound on finite-sample accuracy.  $\square$

This does **not** by itself imply near-perfect or even moderately good finite-sample classification — the projected directions can be arbitrarily small relative to noise. The correct finite-sample benchmark uses the *whitened* mean separation  $d$  of Definition D11, not either mean offset scaled by noise magnitude alone. Under equal priors, the Bayes-optimal accuracy and AUC are the standard Gaussian-discrimination identities

$$\text{Acc}^* = \Phi(d/2), \quad \text{AUC}^* = \Phi(d/\sqrt{2}),$$

obtained by projecting onto the log-likelihood-ratio score (mean  $\pm d^2/2$ , variance  $d^2$  under each hypothesis). These are standard results from classical signal detection theory (see, e.g., [24], or any standard detection-and-estimation reference such as [25]), not a contribution of this paper, but they provide the correct analytical benchmark against which Section 5’s simulated accuracy/AUC values should be compared, rather than treating “rank 2” alone as a qualitative promise of strong separation.

**A notational precision:**  $Y$  in this paper sometimes denotes the raw observation trajectory and sometimes the engineered summary vector supplied to a classifier — these are different objects with different achievable risks. Writing  $Z = \phi(Y)$  for a feature map applied to the raw trajectory, the correct ordering is  $R_{\text{oracle}} \leq R_Y^* \leq R_Z^* \leq R_{\text{const}}$ , with  $R_Z^* \leq R(\hat{\pi}(Z))$  for any fitted policy  $\hat{\pi}$  — a feature ablation changes  $Z$  and can only move  $R_Z^*$ , it does not change  $Y$ ’s complete observation law or  $R_Y^*$ . Section 6’s discussion of the non-Gaussian ABIDES feature summaries should be read as concerning  $R_Z^*$ , the best achievable risk given the engineered features actually used, not  $R_Y^*$ , the best achievable risk given the full raw trajectory — these coincide only when  $\phi$  is sufficient for the classification problem, which is not established for any feature set used in this paper.

**Proposition 1 (Cost of ambiguity is bounded below under confounding — corrected).** The assumptions  $\text{DISRUPT} \neq 0$  and  $\text{HARM} \neq C_{OP}$  alone do not give a cost floor:  $p = 1/2$ ,  $C_{OP} = 3$ ,  $\text{DISRUPT} = 6$ ,  $\text{HARM} = 1$  satisfies both, yet leaving alone is cost-optimal for both true causes ( $\text{HARM} = 1 < C_{OP} = 3$ ), so the ambiguity cost is exactly zero. The proposition therefore also requires that harm strictly exceed the containment cost.

Suppose  $(*)$  holds (the confounded case),  $0 < p < 1$ ,  $\text{DISRUPT} > 0$ , and  $\text{HARM} > C_{OP}$  (harm strictly worse than routine containment — the assumption the earlier version omitted). Then the best achievable expected cost among policies mapping  $\{y_t\}$  to  $\{\text{contain, leave alone}\}$  is

$$R^* = \min\{p C_{OP} + (1 - p) \text{DISRUPT}, \quad p \text{HARM}\},$$

the oracle’s expected cost is  $R_{\text{oracle}} = p C_{OP}$ , and

|                                                                                                                |
|----------------------------------------------------------------------------------------------------------------|
| $\text{CoA}^* = R^* - R_{\text{oracle}} = \min\{(1 - p) \text{DISRUPT}, \quad p (\text{HARM} - C_{OP})\} > 0.$ |
|----------------------------------------------------------------------------------------------------------------|

*Proposition 1 uses* Definition D8 (this paper’s symbols), Definition D9

and, through Theorem 1 (to pass from  $(*)$  to  $P_K^W = P_F^W$ ), Definition D1, Definition D3, Definition D4 and assumptions (A1)–(A5).

*Proof.* Use this paper’s loss table (Definition D8):  $L(\text{contain}, K) = C_{OP}$ ,  $L(\text{contain}, F) = \text{DISRUPT}$ ,  $L(\text{leave alone}, K) = \text{HARM}$ ,  $L(\text{leave alone}, F) = 0$ , and the decision problem of Definition D9 on the window  $W$  of Theorem 1.

*Best achievable cost.* By  $(*)$  and Theorem 1,  $P_K^W = P_F^W =: P$ . For a policy  $\pi$  (randomized allowed, Definition D9) let  $r = \int \pi dP \in [0, 1]$ . Since both expectations in  $R(\pi)$  are taken under the same law  $P$ ,

$$R(\pi) = r [p C_{OP} + (1 - p) \text{DISRUPT}] + (1 - r) p \text{HARM},$$

an affine function of  $r$ . Its infimum over  $r \in [0, 1]$  is attained at an endpoint, and both endpoints are attained by constant deterministic policies ( $\pi \equiv 1$ : always contain;  $\pi \equiv 0$ : always leave alone). Hence

$$R^* = \min\{p C_{OP} + (1 - p) \text{DISRUPT}, p \text{HARM}\},$$

and the value is the same whether the policy class is deterministic (as in the statement) or randomized (as in Definition D9).

*Oracle.* By Definition D9,  $R_{\text{oracle}} = p \min\{C_{OP}, \text{HARM}\} + (1 - p) \min\{\text{DISRUPT}, 0\} = p C_{OP}$ , using  $\text{HARM} > C_{OP}$  and  $\text{DISRUPT} > 0$ .

*Cost of ambiguity.* Subtracting  $p C_{OP}$  inside the minimum,

$$\text{CoA}^* = \min\{(1 - p) \text{DISRUPT}, p (\text{HARM} - C_{OP})\}.$$

Both terms are strictly positive because  $0 < p < 1$ ,  $\text{DISRUPT} > 0$  and  $\text{HARM} > C_{OP}$ ; hence  $\text{CoA}^* > 0$ .  $\square$

For a general finite action/cause loss table  $L(a, M)$ , the corresponding general expression is  $\min_a \{p L(a, K) + (1 - p) L(a, F)\} - p \min_a L(a, K) - (1 - p) \min_a L(a, F)$ , which is strictly positive exactly when no single action is simultaneously optimal for both causes at positive probability — the property Proposition 1’s added assumption ( $\text{HARM} > C_{OP}$ , together with  $\text{DISRUPT} > 0$ ) is what guarantees in the two-action case.

**A more general, non-exact-equality version of this result** (needed to make the decision analysis meaningful under approximate rather than exact confounding, since exact equality is a useful theoretical control case but a brittle operational target). Pointwise minimization of the density-weighted regret gives the following.

**Proposition 2 (Total-variation extension of Proposition 1).** Let the loss table and prior be as in Proposition 1, with  $0 < p < 1$ ,  $\text{DISRUPT} > 0$  and  $\text{HARM} > C_{OP}$ , but do not assume (\*). Let  $f_K, f_F$  be the observation densities (Definition D10) and set  $a = p(\text{HARM} - C_{OP})$ ,  $b = (1 - p)\text{DISRUPT}$ . Then

$$\text{CoA}^* = \int \min\{af_K(y), bf_F(y)\} dy = \frac{a + b - \int |af_K - bf_F| dy}{2},$$

$$\min(a, b) (1 - \text{TV}(P_K, P_F)) \leq \text{CoA}^* \leq \min(a, b),$$

and, if a common surrogate law  $\bar{P}$  satisfies  $\text{TV}(P_K, \bar{P}) \leq \epsilon_K$  and  $\text{TV}(P_F, \bar{P}) \leq \epsilon_F$ , then

$$\text{CoA}^* \geq \min(a, b) \max\{0, 1 - \epsilon_K - \epsilon_F\}.$$

*Proposition 2 uses* Definition D8 (this paper's symbols), Definition D9 and Definition D10; it uses none of (A1)–(A6): it holds for any two observation laws.

*Proof.* Let  $P_K, P_F$  be the two observation laws on the window  $W$ ,  $\mu = P_K + P_F$ , and  $f_K, f_F$  their densities with respect to  $\mu$  (Definition D10; they exist by Radon–Nikodym). Use this paper's table (Definition D8) and the decision problem of Definition D9, with  $a = p(\text{HARM} - C_{OP}) > 0$  and  $b = (1 - p)\text{DISRUPT} > 0$ .

*Step 1 (regret form).* As in Proposition 1,  $\text{HARM} > C_{OP}$  and  $\text{DISRUPT} > 0$  give  $R_{\text{oracle}} = p C_{OP}$ . For any policy  $\pi$ ,

$$R(\pi) = \int \left( \pi [p C_{OP} f_K + (1 - p) \text{DISRUPT} f_F] + (1 - \pi) p \text{HARM} f_K \right) d\mu,$$

and, since  $\int f_K d\mu = 1$ ,

$$R(\pi) - R_{\text{oracle}} = \int \left( \pi b f_F + (1 - \pi) a f_K \right) d\mu.$$

*Step 2 (pointwise minimization).* For  $\pi(y) \in [0, 1]$  the integrand is a convex combination of  $bf_F(y)$  and  $af_K(y)$ , hence  $\geq \min\{af_K(y), bf_F(y)\}$ ; the measurable deterministic policy  $\pi^* = \mathbb{1}[af_K > bf_F]$  attains this pointwise. Therefore

$$\text{CoA}^* = R^* - R_{\text{oracle}} = \int \min\{af_K, bf_F\} d\mu.$$

With  $\min\{x, y\} = \frac{1}{2}(x + y - |x - y|)$ ,  $\int af_K d\mu = a$  and  $\int bf_F d\mu = b$ , this equals  $\frac{1}{2}(a + b - \int |af_K - bf_F| d\mu)$ .

*Step 3 (total variation facts).* For densities  $f, f'$  of  $P, P'$ : with  $B^* = \{f > f'\}$ ,  $P(B^*) - P'(B^*) = \int (f - f')^+ d\mu = \frac{1}{2} \int |f - f'| d\mu$  because  $\int (f - f') d\mu = 0$ ; and for every measurable  $B$ ,  $|P(B) - P'(B)| \leq \max\{\int (f - f')^+ d\mu, \int (f' - f)^+ d\mu\} = \frac{1}{2} \int |f - f'| d\mu$ . This is the equality in Definition D10, and it gives  $1 - \text{TV}(P, P') = \int \min\{f, f'\} d\mu$ . From the supremum form, TV satisfies the triangle inequality:  $|P(B) - P''(B)| \leq |P(B) - P'(B)| + |P'(B) - P''(B)|$  for every  $B$ .

*Step 4 (bounds).* Upper:  $\int \min\{af_K, bf_F\} \leq \min\{\int af_K, \int bf_F\} = \min(a, b)$ . Lower:  $af_K \geq \min(a, b)f_K$  and  $bf_F \geq \min(a, b)f_F$ , so  $\min\{af_K, bf_F\} \geq \min(a, b) \min\{f_K, f_F\}$ , and integrating with Step 3 gives  $\text{CoA}^* \geq \min(a, b)(1 - \text{TV}(P_K, P_F))$ .

*Step 5 (surrogate).* If  $\text{TV}(P_K, \bar{P}) \leq \epsilon_K$  and  $\text{TV}(P_F, \bar{P}) \leq \epsilon_F$ , the triangle inequality gives  $\text{TV}(P_K, P_F) \leq \epsilon_K + \epsilon_F$ , so Step 4 gives  $\text{CoA}^* \geq \min(a, b)(1 - \epsilon_K - \epsilon_F)$ ; and  $\text{CoA}^* \geq 0$  as an integral of a nonnegative function. Hence  $\text{CoA}^* \geq \min(a, b) \max\{0, 1 - \epsilon_K - \epsilon_F\}$ .

None of the model assumptions (A1)–(A6) is used: the argument holds for any two observation laws.  $\square$

The first identity connects to the classical relationship between Bayes risk and total variation distance in binary hypothesis testing (see, e.g., [26], or the Scheffé identity underlying it). The lower bound recovers the exact-confounding Proposition 1 when  $\text{TV} = 0$ . The surrogate-law bound states precisely the extra premise any “approximate confounding transfers approximately” argument needs: a controlled bound on distributional approximation error, not merely a match on one summary statistic. This is the correct standard against which Section 7’s ABIDES scenarios should be judged, and one important empirical limit applies: **a classifier performing near chance cannot certify a small total variation distance** — failure of specific tested classifiers is evidence of a ceiling on what *those* methods extracted, not proof of an upper bound on total variation or on what any classifier could extract.

This result is the formal content behind the paper’s title: *ambiguity has a cost that no amount of passive observation, however cleverly analyzed, can eliminate* — a limit, not merely an empirical shortfall of current classifiers.

**An unequal-prior extension of the Gaussian benchmark that follows Corollary 2, completing the decision-cost analog of Corollary**

**2's discriminability formulas.** For the general loss table of Definition D8 applied to the equal-covariance Gaussian raw-trajectory setting above, direct likelihood-ratio integration gives the population-optimal ambiguity cost as a function of discriminability  $d$ .

**Proposition 3 (Gaussian unequal-prior ambiguity cost).** Let  $Y \mid M \sim \mathcal{N}(\mu_M, V)$ ,  $M \in \{K, F\}$ , with common covariance  $V \succ 0$  and discriminability  $d > 0$  (Definition D11), prior  $p = P(K)$ , and the general loss table with  $a = p(H_K - C_K)$ ,  $b = (1 - p)(C_F - H_F)$ ,  $a, b > 0$ . Then the population-optimal ambiguity cost is

$$\text{CoA}_Y^*(d) = a \Phi\left(\frac{\log(b/a)}{d} - \frac{d}{2}\right) + b \Phi\left(\frac{\log(a/b)}{d} - \frac{d}{2}\right),$$

which tends to  $\min(a, b)$  as  $d \rightarrow 0^+$  (the value at  $d = 0$ , exact confounding, recovering Proposition 1, is obtained as this limit) and decreases toward 0 as  $d \rightarrow \infty$ .

*Proposition 3 uses* Definition D8 (general symbols), Definition D9, Definition D10 and Definition D11; applied to the model on a finite window it needs (A1)–(A5) (common covariance) and (A6) ( $V \succ 0$ ).

*Proof.* Use the general loss table (Definition D8) and Definition D9 with observation  $Y$ . Here  $a = p(H_K - C_K) > 0$  and  $b = (1 - p)(C_F - H_F) > 0$  force  $0 < p < 1$ ,  $C_K < H_K$  and  $H_F < C_F$ .

*Step 1 (regret form).* The oracle contains under  $K$  and leaves alone under  $F$ , so  $R_{\text{oracle}} = p C_K + (1 - p) H_F$  (Definition D9). Exactly as in Steps 1–2 of the proof of the total-variation extension of Proposition 1, now with the general table,

$$R(\pi) - R_{\text{oracle}} = \int (\pi b f_F + (1 - \pi) a f_K) dy, \quad \text{CoA}_Y^* = \int \min\{a f_K, b f_F\} dy,$$

attained by  $\pi^* = \mathbb{1}[a f_K > b f_F]$ , where  $f_M$  is the  $\mathcal{N}(\mu_M, V)$  density (Lebesgue measure dominates both laws since  $V \succ 0$ ; Definition D10). Splitting the integral on the event  $\{a f_K \leq b f_F\}$ ,

$$\text{CoA}_Y^* = a P_K(a f_K(Y) \leq b f_F(Y)) + b P_F(a f_K(Y) > b f_F(Y)).$$

*Step 2 (log-likelihood ratio).* With  $\delta = \mu_K - \mu_F$  and  $d = \sqrt{\delta^\top V^{-1} \delta}$  (Definition D11),

$$\Lambda(y) = \log \frac{f_K(y)}{f_F(y)} = \delta^\top V^{-1} \left( y - \frac{1}{2}(\mu_K + \mu_F) \right),$$

because the quadratic terms in  $y$  cancel (common  $V$ ).  $\Lambda(Y)$  is an affine function of a Gaussian vector: under  $K$  its mean is  $\frac{1}{2}\delta^\top V^{-1}\delta = d^2/2$ , under  $F$  it is  $-d^2/2$ , and under both its variance is  $\delta^\top V^{-1}VV^{-1}\delta = d^2$ . Since  $V \succ 0$ ,  $d > 0$  iff  $\delta \neq 0$ ; with  $d > 0$  these laws are non-degenerate, so  $\{\Lambda = c\}$  is null under both.

*Step 3 (closed form).* Let  $c = \log(b/a)$ , well defined since  $a, b > 0$ . Then  $af_K \leq bf_F \iff \Lambda \leq c$ , and

$$P_K(\Lambda \leq c) = \Phi\left(\frac{c - d^2/2}{d}\right) = \Phi\left(\frac{\log(b/a)}{d} - \frac{d}{2}\right),$$

$$P_F(\Lambda > c) = 1 - \Phi\left(\frac{c + d^2/2}{d}\right) = \Phi\left(\frac{\log(a/b)}{d} - \frac{d}{2}\right),$$

using  $1 - \Phi(x) = \Phi(-x)$ . Substituting in Step 1 gives the stated formula; the division by  $d$  is why  $d > 0$  is required.

*Step 4 (monotonicity).* Write  $g(d)$  for the right-hand side,  $\alpha = c/d - d/2$ ,  $\beta = -c/d - d/2$ , and  $\varphi = \Phi'$ . Since  $\alpha^2 - \beta^2 = (\alpha - \beta)(\alpha + \beta) = (2c/d)(-d) = -2c$ ,  $\varphi(\alpha)/\varphi(\beta) = e^c = b/a$ , i.e.  $a\varphi(\alpha) = b\varphi(\beta)$ . Hence

$$g'(d) = a\varphi(\alpha)\left(-\frac{c}{d^2} - \frac{1}{2}\right) + b\varphi(\beta)\left(\frac{c}{d^2} - \frac{1}{2}\right) = -a\varphi(\alpha) < 0,$$

so  $g$  is strictly decreasing on  $(0, \infty)$ .

*Step 5 (limits).* As  $d \rightarrow \infty$ ,  $\alpha, \beta \rightarrow -\infty$ , so  $g(d) \rightarrow 0$ . As  $d \rightarrow 0^+$ : if  $a < b$  then  $c > 0$ ,  $\alpha \rightarrow +\infty$ ,  $\beta \rightarrow -\infty$  and  $g \rightarrow a$ ; if  $a > b$  then  $c < 0$ ,  $\alpha \rightarrow -\infty$ ,  $\beta \rightarrow +\infty$  and  $g \rightarrow b$ ; if  $a = b$  then  $c = 0$ ,  $\alpha = \beta = -d/2 \rightarrow 0$  and  $g \rightarrow (a+b)/2 = a$ . In all cases  $g(d) \rightarrow \min(a, b)$ . At  $d = 0$  ( $\delta = 0$ ,  $P_K = P_F$ , exact confounding) Step 1 gives  $\text{CoA}_Y^* = f \min\{af, bf\} = \min(a, b)$  directly, so the limit equals the  $d = 0$  value; in this paper's symbols (Definition D8) this is Proposition 1's  $\min\{(1-p)\text{DISRUPT}, p(\text{HARM} - C_{OP})\}$ .

Applying the proposition to the model on a finite window requires the stacked observations to be Gaussian with a common covariance, which assumptions (A1)–(A5) give as in Steps 1–2 of the proofs of Theorems 1 and 2, and  $V \succ 0$ , which (A6) gives; the proof above does not otherwise use the model.  $\square$

This is a standard consequence of Gaussian hypothesis testing, not a new result, but it supplies the missing unequal-prior cost benchmark for the raw-trajectory case; comparing any of this paper's summary-based classifiers against it should be done while explicitly acknowledging that a summary

statistic is an information-restricted version of the raw trajectory, so  $R_Z^* \geq \text{CoA}_Y^* + R_{\text{oracle}}$  in general, not equality.

#### 4.1. *Confounding is fragile even within the linear model: a state-feedback extension*

Theorem 1’s confounding condition has a hidden assumption worth making explicit: the incident is modeled as an **output-level** (sensor-level) disturbance — it shifts  $x_t$  only at the point of observation ( $\tilde{x}_t = x_t + \theta_M b_M$ ), while the state’s own recursion ( $x_{t+1} = Ax_t + w_t$ ) evolves as if the incident never happened. This is a real simplification. A cyber compromise or a persistent order-generation fault plausibly behaves more like a **state-level** (process) disturbance: the perturbed behavior at time  $t$  feeds into what the system does at  $t + 1$ , not just into what is observed at  $t$  — exactly how an incident agent’s behavior in a real order book feeds into subsequent book state, which other agents then react to.

We show that this distinction is not cosmetic: replacing the output-level model by the state-level model below (the perturbation propagates) makes Theorem 1’s confounding condition dramatically more restrictive, and generically fails to hold for exactly the  $(b_K, b_F)$  pairs Corollary 1 constructs.

**Definition D2 (State-level injection model).** As Definition D1, except that the perturbed state drives the recursion from the onset on:

$$\begin{aligned} x_{t+1} &= Ax_t + w_t \quad (t < \tau), & x_{t+1} &= A\tilde{x}_t + w_t \quad (t \geq \tau), \\ \tilde{x}_t &= x_t + \mathbb{K}[t \geq \tau] \theta_M b_M, & y_t &= C\tilde{x}_t + \varepsilon_t. \end{aligned}$$

The injection is persistent (added at every  $t \geq \tau$ ) and propagates through  $A$ .

**Definition D6 (Unobservable subspace; observability).** The (Kalman) unobservable subspace of  $(A, C)$  is  $\mathcal{N}(A, C) = \bigcap_{k=0}^{n-1} \ker(CA^k)$ . The pair  $(A, C)$  is observable if  $\mathcal{N}(A, C) = \{0\}$ , equivalently if the stacked matrix

$$\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

has rank  $n$ .

**Theorem 2 (State-feedback confounding requires unobservability).** Under the state-level injection model, let  $\Delta_0 = \theta_K b_K - \theta_F b_F$ . Mecha-

nism instances  $(K, \theta_K, b_K)$  and  $(F, \theta_F, b_F)$  are observationally indistinguishable at every  $t \geq \tau$  if and only if  $\Delta_0$  lies in the unobservable subspace of the pair  $(A, C)$ , i.e.

$$\Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k),$$

the classical (Kalman) unobservable subspace of  $(A, C)$ , where  $n$  is the state dimension.

*Theorem 2 uses* Definition D2, Definition D3, Definition D4 (window  $W = \{t : t \geq \tau\}$ ), Definition D6 and assumptions (A1)–(A4) with (A5) in its all-time form.

*Proof.* Work in the state-level injection model (Definition D2) with window  $W = \{t : t \geq \tau\}$  (Definition D4); by (A4) the onset  $\tau$  is deterministic and common to both instances (Definition D3). Write  $S_k = \sum_{j=0}^k A^j$  for  $k \geq 0$ .

*Step 1 (noise part and deterministic part).* Define the uninjected process  $z_0 = x_0$ ,  $z_{t+1} = Az_t + w_t$  for all  $t \geq 0$ , and the deterministic sequence  $s_t = 0$  for  $t \leq \tau$ ,  $s_{t+1} = As_t + A\theta_M b_M$  for  $t \geq \tau$ . Then  $x_t = z_t + s_t$  for all  $t$ : this holds for  $t \leq \tau$  because the recursion before the onset is uninjected, and if it holds at some  $t \geq \tau$ , then by Definition D2

$$x_{t+1} = A(x_t + \theta_M b_M) + w_t = (Az_t + w_t) + (As_t + A\theta_M b_M) = z_{t+1} + s_{t+1}.$$

Unrolling,  $s_{\tau+k} = \sum_{j=1}^k A^j \theta_M b_M$ , so  $\tilde{x}_{\tau+k} = z_{\tau+k} + S_k \theta_M b_M$  and, for every  $k \geq 0$ ,

$$y_{\tau+k} = G_{\tau+k} + m_M(\tau + k), \quad G_t = Cz_t + \varepsilon_t, \quad m_M(\tau + k) = CS_k \theta_M b_M.$$

*Step 2 (why the noise does not matter).*  $G = (G_t)_{t \geq 0}$  is a fixed measurable function of  $(x_0, \{w_t\}, \{\varepsilon_t\})$  through  $A$  and  $C$  and does not involve  $\tau$  or the instance. By (A1), (A2) and (A3) its law, in particular its covariance, is the same under both instances. The instance enters only through the deterministic mean shift  $m_M$ .

*Step 3 (laws equal iff shifts equal).* If  $m_K(t) = m_F(t)$  for all  $t \in W$ , then  $Y_W = G_W + m$  under both instances, so every finite-dimensional marginal of  $Y_W$  is the same translate of the common marginal of  $G$ , and  $P_K^W = P_F^W$  (Definition D4). Conversely, if  $P_K^W = P_F^W$ , then since each  $G_t$  is Gaussian by (A1) and (A2), and hence integrable,  $\mathbb{E}_K[y_t] = \mathbb{E}_F[y_t]$  and so  $m_K(t) = m_F(t)$

for every  $t \in W$ . With  $\Delta_0 = \theta_K b_K - \theta_F b_F$ ,  $m_K(\tau + k) - m_F(\tau + k) = CS_k \Delta_0$ . Hence

$$P_K^W = P_F^W \iff CS_k \Delta_0 = 0 \text{ for all } k \geq 0.$$

*Step 4 (telescoping).* For every  $N \geq 0$ :  $CS_k \Delta_0 = 0$  for  $k = 0, \dots, N \iff CA^k \Delta_0 = 0$  for  $k = 0, \dots, N$ . Indeed  $CS_0 = C$  and  $CS_k - CS_{k-1} = CA^k$  for  $k \geq 1$ , so each family is obtained from the other by taking successive differences or partial sums. Hence  $P_K^W = P_F^W \iff CA^k \Delta_0 = 0$  for all  $k \geq 0$ .

*Step 5 (Cayley–Hamilton).* Let  $\chi(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_0$  be the characteristic polynomial of  $A$ . By the Cayley–Hamilton theorem  $A^n = -\sum_{i=0}^{n-1} c_i A^i$ ; multiplying by  $A^{k-n}$  and inducting on  $k$ , every  $A^k$  with  $k \geq n$  is a linear combination  $\sum_{i=0}^{n-1} \alpha_{k,i} A^i$ . Therefore, if  $CA^i \Delta_0 = 0$  for  $i = 0, \dots, n-1$ , then  $CA^k \Delta_0 = \sum_i \alpha_{k,i} CA^i \Delta_0 = 0$  for every  $k \geq n$ ; the converse is immediate. So

$$P_K^W = P_F^W \iff \Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k) = \mathcal{N}(A, C)$$

(Definition D6), which is the claim.

*Remark.* Adding pre-onset times to  $W$  changes nothing, since  $m_K(t) = m_F(t) = 0$  for  $t < \tau$ .  $\square$

**A finite-horizon statement, making explicit that this result does not require infinite data to characterize.**

**Theorem 3 (Finite-horizon version of Theorem 2).** Under the state-level injection model, let  $\Delta_0 = \theta_K b_K - \theta_F b_F$ , let  $H \geq 1$  be the number of post-onset observations, and define

$$G_H = \begin{bmatrix} C \\ C(I + A) \\ \vdots \\ C \sum_{j=0}^{H-1} A^j \end{bmatrix}.$$

Mechanism instances  $(K, \theta_K, b_K)$  and  $(F, \theta_F, b_F)$  induce the same law of the observations over the window of  $H$  post-onset observations if and only if  $G_H \Delta_0 = 0$ , which holds if and only if  $CA^j \Delta_0 = 0$  for  $j = 0, \dots, H-1$ . If  $H \geq n$ , this is equivalent to the all-time condition of Theorem 2,  $\Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k)$ .

*Theorem 3* uses Definition D2, Definition D3, Definition D4 (window  $W = \{\tau, \dots, \tau + H - 1\}$ ), Definition D6 and assumptions (A1)–(A4) with (A5) in its finite-horizon form.

*Proof.* Take  $W = \{\tau, \dots, \tau + H - 1\}$  (Definition D4) and the notation of the proof of Theorem 2 (Definition D2, Definition D3, (A1)–(A4)). Steps 1–3 of that proof use only the times in  $W$ , so they give

$$P_K^W = P_F^W \iff CS_k \Delta_0 = 0 \text{ for } k = 0, \dots, H - 1, \quad S_k = \sum_{j=0}^k A^j.$$

The block rows of  $G_H$  are exactly  $CS_0, CS_1, \dots, CS_{H-1}$ , so this is  $G_H \Delta_0 = 0$ . By Step 4 of that proof with  $N = H - 1$ , it is equivalent to  $CA^j \Delta_0 = 0$  for  $j = 0, \dots, H - 1$ .

If  $H \geq n$ , these conditions include  $j = 0, \dots, n - 1$ , which by Step 5 (Cayley–Hamilton) imply  $CA^j \Delta_0 = 0$  for all  $j \geq 0$ ; conversely the all-time condition implies the conditions for  $j < H$ . Hence for  $H \geq n$  the finite-window condition is equivalent to  $\Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k)$  (Definition D6). For  $H < n$  the finite-window condition is in general strictly weaker.

As in Theorem 2, the noise enters only through the law of  $G_W$ , which does not depend on the instance, so the result holds for every noise level allowed by (A1); it concerns equality of laws, not finite-sample discriminability.  $\square$

**The result is therefore not inherently asymptotic:** exact equality over any window of at least  $n$  post-onset observations already forces the all-time condition; what remains genuinely dependent on horizon and noise level is *practical, finite-sample discriminability* when the condition fails, which Theorem 2’s population-level statement does not by itself bound (see qualification 3 below).

**Corollary 3 (State-feedback confounding under observability).** Under the hypotheses of Theorem 2, if  $(A, C)$  is observable, then mechanism instances  $(K, \theta_K, b_K)$  and  $(F, \theta_F, b_F)$  are observationally indistinguishable at every  $t \geq \tau$  if and only if  $\theta_K b_K = \theta_F b_F$ .

*Corollary 3* uses everything Theorem 2 uses, and Definition D6; its accompanying remark also uses Definition D7.

*Proof.* By Theorem 2 (Definition D2, Definition D3, Definition D4, (A1)–(A4), (A5) in its all-time form), the instances are indistinguishable at every  $t \geq \tau$  iff  $\Delta_0 = \theta_K b_K - \theta_F b_F \in \mathcal{N}(A, C)$ . If  $(A, C)$  is observable,  $\mathcal{N}(A, C) = \{0\}$  (Definition D6), so the condition is  $\Delta_0 = 0$ , i.e.  $\theta_K b_K = \theta_F b_F$ .  $\square$

**Definition D7 (Generic).** For fixed  $C$ , a property of  $A$  holds generically if it holds for Lebesgue-almost every  $A \in \mathbb{R}^{n \times n}$ , i.e. it fails only on a Lebesgue-null subset of  $\mathbb{R}^{n \times n}$ .

**Remark (Theorem 1’s confounded pairs are generically state-feedback-fragile).** Theorem 1 only requires  $\Delta_0 \in \ker(C)$  — a single condition. Theorem 2 requires  $\Delta_0$  to survive  $n$  such conditions simultaneously ( $\ker(C), \ker(CA), \dots, \ker(CA^{n-1})$ ). Whenever  $(A, C)$  is observable (for fixed  $C \neq 0$ , the generic case for a randomly chosen  $A$ ), the unobservable subspace is  $\{0\}$ , so **the only state-feedback-robust confounding is the trivial one where  $\theta_K b_K = \theta_F b_F$  exactly as vectors** — i.e., the two mechanisms must push the identical linear combination of latent state variables, not merely produce the identical *projection* under  $C$ .

*Proof. (i) One condition versus  $n$ .* For the output-level model (Definition D1),  $(*)$  reads  $C(\theta_K b_K - \theta_F b_F) = 0$ , i.e.  $\Delta_0 \in \ker(C)$ . For the state-level model (Definition D2), Theorem 2 requires  $\Delta_0 \in \ker(CA^k)$  for  $k = 0, \dots, n-1$ .

*(ii) Observable implies trivial robust confounding.* This is the precise Corollary 3: if  $(A, C)$  is observable then  $\mathcal{N}(A, C) = \{0\}$  and state-feedback-robust confounding requires  $\theta_K b_K = \theta_F b_F$  as vectors.

*(iii) Genericity for fixed  $C \neq 0$*  (Definition D7). Let  $\mathcal{O}(A) = [C; CA; \dots; CA^{n-1}] \in \mathbb{R}^{nm \times n}$ . The set  $U = \{A : \text{rank } \mathcal{O}(A) < n\}$  of  $A$  for which  $(A, C)$  is not observable (Definition D6) is the common zero set of the  $n \times n$  minors of  $\mathcal{O}(A)$ , each a polynomial in the entries of  $A$ .

*A witness.* Since  $C \neq 0$ , it has a nonzero row  $c^\top$ . Extend  $v_1 = c^\top$  to a basis  $v_1, \dots, v_n$  of row vectors of  $\mathbb{R}^{1 \times n}$  and define  $A_0$  by  $v_k A_0 = v_{k+1}$  ( $k < n$ ),  $v_n A_0 = 0$ ; this determines  $A_0$  uniquely since the  $v_k$  form a basis. Then  $c^\top A_0^k = v_{k+1}$  for  $k = 0, \dots, n-1$ ; these  $n$  rows of  $\mathcal{O}(A_0)$  are linearly independent, so the minor  $D(A)$  formed by the corresponding rows of  $\mathcal{O}(A)$  satisfies  $D(A_0) \neq 0$ .

*Null set.*  $D$  is a polynomial that is not identically zero, so its zero set is Lebesgue-null in  $\mathbb{R}^{n \times n}$  (standard: by induction on the number of variables and Fubini). Since  $U \subseteq \{D = 0\}$ ,  $U$  is null, so observability holds for Lebesgue-almost every  $A$ ; equivalently it holds with probability one when  $A$  is drawn from any law absolutely continuous with respect to Lebesgue measure.

*$C \neq 0$  is needed.* If  $C = 0$ ,  $\mathcal{O}(A) = 0$  for every  $A$ , so no  $A$  gives an observable pair ( $n \geq 1$ ).

*Not derived.* Genericity in the sense of Definition D7 says nothing about

which  $A$  arise in deployed markets; the corrected remark therefore makes no claim about “realistically chosen”  $A$ .  $\square$

Theorem 1’s construction (Section 5, using  $C$ ’s null space to make  $b_K \neq b_F$  while  $Cb_K = Cb_F$ ) is exactly the kind of pair, with  $\Delta_0 \in \ker(C)$  but  $\theta_K b_K \neq \theta_F b_F$ , that Corollary 3 says cannot remain confounded under state-level feedback whenever  $(A, C)$  is observable; by the remark, for fixed  $C \neq 0$  this is the case for Lebesgue-almost every  $A$ . Whether a given market’s dynamics are observable in this sense is not derived here (qualification 2 below).

**Empirical confirmation.** Using the exact  $(A, C, b_K, b_F)$  from Section 5’s pilot, with  $A = 0.6I$  (a scalar multiple of the identity), classification stays at chance across the injection magnitudes tested even under state-level feedback. **The relevant fact here is simpler than commutativity in general** (a claim that only makes sense for compatible square matrices, and is not the precise mechanism at work): since  $A = 0.6I$ , we have  $CA^k = 0.6^k C$  for every  $k$ , so  $\ker(C)$  is trivially  $A$ -invariant, making this a non-generic, measure-zero special case where Theorem 1’s weaker condition happens to already satisfy Theorem 2’s stronger one. Replacing  $A$  with a genuinely coupled matrix (off-diagonal terms, stable, confirmed fully observable with  $(A, C)$  via  $\text{rank}[C; CA; CA^2] = 3$ ) and re-running the identical experiment: under state-level injection with generic coupled dynamics, classification accuracy rises with injection magnitude, while under output-level injection it stays at chance throughout; with scalar (uncoupled) dynamics the state-level case also stays at chance. **The same  $(b_K, b_F)$  pair that is exactly confounded under the output-level model becomes classifiable, increasingly so with injection magnitude, once the identical perturbation propagates through coupled dynamics.** These classifiers use features computed over the whole episode; how quickly the difference becomes detectable after onset is not measured here (qualification 3 below).

**Four qualifications on this result:** 1. **Repeated injection is not required for the qualitative phenomenon, though the precise mean-difference sequence differs.** A single, one-time initial-state perturbation produces the sequence of mean differences  $CA^k \Delta_0$  at each step  $k$ , whereas persistent forcing produces the *cumulative sum*  $C \sum_{j=0}^k A^j \Delta_0$  — these are different sequences. **What the two cases do share is the identical all-time unobservability condition:** if  $CA^j \Delta_0 = 0$  for every  $j = 0, \dots, n-1$  (which, by Cayley-Hamilton, extends to every  $j \geq 0$ ), then every term in *both*

sequences vanishes, so the qualitative phenomenon (confounding surviving feedback) requires the same condition either way — the persistent, repeated injection used in Section 3.2 is a modeling choice affecting the specific trajectory of mean differences, not a necessary ingredient of the unobservability condition itself.

2. **“Generic” does not mean “typical of deployed markets.”** For fixed  $C \neq 0$ , an  $A$  drawn from any distribution with a density is observable with probability 1, but real, structured systems can have unobservable components *by construction* — the claim that realistic market dynamics behave like a generic matrix is an assumption, not a derived fact, and we have not established it.

3. **Exact observability does not imply useful discrimination over a finite, noisy horizon.** For example,  $C = (1, 0)$  and  $A = \begin{pmatrix} 0.6 & \epsilon \\ 0 & 0.6 \end{pmatrix}$  is observable for every  $\epsilon \neq 0$ , yet a hidden difference in the second coordinate becomes arbitrarily hard to detect in finite time as  $\epsilon \rightarrow 0$ . Theorem 2 characterizes an asymptotic, all-time property; it does not by itself bound how many time steps are needed to exploit observability in practice.

4. **Identical state effects can still represent different causes.**  $\Delta_0 = 0$  (the trivial, fully robust case) is mathematically degenerate but operationally significant: a fault and a compromised process can produce literally the same downstream state trajectory while differing in what they represent (authorized versus unauthorized). Observability of the state’s evolution does not by itself identify intent, authorization, or legal status — a distinction the identifiability framework in this paper is not built to make and does not claim to.

**Why this matters for Section 7.** This result requires no nonlinearity, no multi-agent interaction, and no ABIDES at all to demonstrate — it is a fact about the *linear* model itself, once a more realistic injection semantics (state-level rather than output-level) is used. It gives a cleaner, more fundamental candidate explanation for part of Section 7’s instability than feature choice or incident footprint alone: a real incident’s behavioral change plausibly propagates through system state (order flow affects the book, which affects future order flow), which is structurally a state-level disturbance, and Corollary 3 says such disturbances are confounding-fragile by default, robustly so only in special, non-generic dynamics. Section 7.8 later attempts to test this prediction directly in the ABIDES setting; the initial persistence ablation showed a large effect in the predicted direction, but this design conflates persistence with incident duration and cumulative activity, so it does not cleanly isolate the mechanism, and a follow-up isolation redesign (Section 7.10.2) remains inconclusive at the sample size tested. We therefore

state this section’s contribution at the strength it actually has: a proven, exact mechanism *within the linear model*, directionally corroborated but not yet cleanly isolated in the nonlinear setting.

## 5. Pilot-scale empirical validation

We instantiate the model at small scale ( $n = 3$  latent dimensions,  $m = 2$  observed features,  $T = 40$  steps per episode,  $AR(1)$  dynamics) to confirm the theorem’s predictions computationally before relying on them. This is a **deliberately minimal, synthetic instantiation** — a proof of concept that the theory is internally consistent and empirically detectable in simulation — not a market-microstructure-realistic study; the realistic-market study is Section 7.

Two regimes are constructed by choice of injection directions  $b_K, b_F$ : - **CONFOUNDED**:  $b_K, b_F$  chosen (via the observation matrix’s null space) so that  $Cb_K = Cb_F$  exactly, satisfying (\*). - **SEPARATED**:  $b_K, b_F$  chosen so that  $Cb_K, Cb_F$  are linearly independent.

A logistic-regression classifier is trained on simple trajectory summary statistics (mean, standard deviation, and net drift of each observed feature) to distinguish compromise from fault episodes, in both regimes, at two injection magnitudes.

**Table 1. Passive classifier discrimination, by regime and magnitude (held-out episodes).**

| Regime                                  | Magnitude | Accuracy | AUC    |
|-----------------------------------------|-----------|----------|--------|
| CONFOUNDED (theorem: indistinguishable) | 0.6       | [L7-2]   | [L7-2] |
| SEPARATED (theorem: distinguishable)    | 0.6       | [L7-2]   | [L7-2] |
| CONFOUNDED (theorem: indistinguishable) | 1.2       | [L7-2]   | [L7-2] |
| SEPARATED (theorem: distinguishable)    | 1.2       | [L7-2]   | [L7-2] |

As predicted by Theorem 1 and Proposition 1: in the CONFOUNDED regime, classifier accuracy sits at chance level (one half) **regardless of injection magnitude** — more severe incidents are not easier to classify when the confounding condition holds, exactly because the two hypotheses remain observationally identical at any magnitude satisfying (\*). In the SEPARATED

regime, accuracy is high and *improves further* with magnitude, as Corollary 2 predicts via the effective signal-to-noise ratio.

### 5.1. Robustness to classifier choice

Theorem 1 (equal observation laws) guarantees the chance-level cap holds for *any* passive classifier under confounding, not just logistic regression. We confirm this against three additional, structurally different classifier families — a random forest (a flexible nonparametric model capable of fitting complex nonlinear decision boundaries within the limits of its finite depth and training data, not a literal universal approximator), Gaussian naive Bayes, and a naive single-feature threshold heuristic (mimicking a simple real-world detector) — using the same features and train/test split.

**Table 1b. Classifier robustness check, CONFOUNDED vs. SEPARATED (held-out episodes).**

| Classifier                 | CONFOUNDED<br>accuracy | CONFOUNDED<br>AUC | SEPARATED<br>accuracy | SEPARATED<br>AUC |
|----------------------------|------------------------|-------------------|-----------------------|------------------|
| Logistic Regression        | [L7-2]                 | [L7-2]            | [L7-2]                | [L7-2]           |
| Random Forest              | [L7-2]                 | [L7-2]            | [L7-2]                | [L7-2]           |
| Gaussian Naive Bayes       | [L7-2]                 | [L7-2]            | [L7-2]                | [L7-2]           |
| Threshold Rule (heuristic) | [L7-2]                 | [L7-2]            | [L7-2]                | [L7-2]           |

Every classifier family — including the random forest, which is flexible enough to fit complex nonlinear patterns in the training data if any existed — is pinned at chance level under confounding, and every classifier substantially exceeds chance under separation. This confirms the chance-level result is a property of the confounding condition itself (as Theorem 1 implies, and which holds for *any* classifier, not only the flexible ones tested here), not an artifact of logistic regression’s specific inductive bias. The threshold heuristic is the weakest performer under separation, as expected of a single-feature rule, but still comfortably above chance — showing separation is detectable even by a naive real-world-style detector once the confounding condition does not

hold. (Accuracy in this table uses each classifier’s standard decision boundary at one half, the conventional choice for a classification-quality metric; AUC does not use any single threshold, since it is computed by ranking continuous scores across all possible thresholds. Section 6 below uses the cost-optimal threshold specific to the loss structure, generally different from one half, for the decision-cost analysis.)

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## 6. Cost-of-ambiguity quantification

We attach a cost structure — containment cost  $C_{OP} = 3$ , unresolved-harm cost  $HARM = 10$ , needless-disruption cost  $DISRUPT = 6$  (toy units, satisfying the corrected Proposition 1’s required  $HARM > C_{OP}$  and  $DISRUPT > 0$ ; not calibrated to any real deployment) — and compare policies across a sweep of the prior  $P(K)$ .

**Terminology.** Three distinct risk quantities must be kept separate, in the ordering  $R_{\text{oracle}} \leq R_Y^* \leq R_{\text{const}}$ , where  $R_{\text{oracle}} = \mathbb{E}[\min_a L(a, M)]$  is the true-cause oracle,  $R_Y^* = \inf_{\pi} \mathbb{E}[L(\pi(Y), M)]$  is the best achievable policy using only the observation  $Y$ , and  $R_{\text{const}} = \min_a \mathbb{E}[L(a, M)]$  is the best *constant*-action policy.  $R_Y^*$  and  $R_{\text{const}}$  coincide under exact confounding, but not in general. Exact confounding is a *sufficient*, not necessary, condition for this equality: under  $(*)$ ,  $Y$  carries no information, so  $R_Y^* = R_{\text{const}}$  trivially. But the two can also coincide even when  $Y$  is informative, if the available information is never strong enough to flip the loss-optimal action across the relevant range of the observation — for example, at a low prior  $p$ , a moderately informative  $Y$  can still leave both posteriors  $P(K | Y=0)$  and  $P(K | Y=1)$  below the containment threshold, so the optimal action never changes and  $R_Y^* = R_{\text{const}}$  despite  $Y$  carrying real information. We do not claim to have distinguished these two cases empirically in this paper’s simulations; the point is definitional, to avoid treating an absence of measured cost benefit as proof that the underlying observation laws are equal. Under separation, in our specific simulated instances,  $R_Y^*$  can be — and empirically is — substantially *below*  $R_{\text{const}}$ , so  $R_{\text{const}}$  is not the optimum in that regime, and the tables below label it accordingly.

**Prior-shift correction of the decision threshold.** The classifier is trained on a *balanced* (50/50) dataset, so its raw output  $s(y)$  estimates  $P(K | Y = y)$  under a training prior  $p_{\text{train}} = 0.5$ , not the deployment prior

$p$ . Applying the deployment-calibrated threshold  $q^* = \text{DISRUPT}/(\text{HARM} - C_{OP} + \text{DISRUPT})$  directly to this raw score is a prior-shift error of exactly the kind documented for cost-sensitive learning [22, Sec. 3, Thm. 2]; see also [23]. The effect is not minor: at the lowest prior of the grid, the deployment threshold  $q^*$  lies below one half, while the correctly prior-adjusted threshold  $s^*$  on a balanced-training score lies far above it — since an uninformative classifier’s raw score hovers near one half, comparing it to  $q^*$  makes it mostly predict “contain,” while the correct comparison point  $s^*$  makes it mostly and correctly predict “leave.” **With the standard Bayes’ rule prior-correction**  $q_p(y) = \frac{(p/p_{\text{train}})^{s(y)}}{(p/p_{\text{train}})^{s(y)} + \frac{1-p}{1-p_{\text{train}}}(1-s(y))}$ , **our fitted classifier’s empirical cost under confounding is statistically indistinguishable from  $R_{\text{const}}$  (to within simulation noise); without it, the same classifier appears actively worse than ignoring the observation, an artifact of the threshold rather than of confounding.** We are precise about what this does and does not establish: Proposition 1 characterizes the Bayes-optimal policy using  $Y$  ( $R_Y^*$ ) as equal to  $R_{\text{const}}$  under exact confounding — it does not guarantee that any *given fitted* policy will exactly attain that value, since a learned score can retain residual estimation or calibration error even after correct prior adjustment, and such error can in principle still produce a policy strictly worse than  $R_{\text{const}}$ . What we have shown is empirical: our specific corrected classifier approximately attained the theoretical floor in this experiment, not that every classifier must.

We compare four policies: **Fixed** (always contain), **Leave** (always leave alone), **Classifier** (the passive classifier’s prior-corrected posterior, thresholded at  $q^*$ ), and **Oracle**.

**Table 2. Expected cost per episode (toy units), CONFOUNDED regime.**

| $P(K)$ | Classifier |         |                   |         | $R_{\text{const}}$ | Classifier  |         |
|--------|------------|---------|-------------------|---------|--------------------|-------------|---------|
|        | Fixed      | Leave   | (prior-corrected) | Oracle  | (best constant)    | excess over | $R_Y^*$ |
| 0.05   | [L7-3a]    | [L7-3a] | [L7-3a]           | [L7-3a] | [L7-3a]            | [L7-3a]     | [L7-3a] |
| 0.10   | [L7-3a]    | [L7-3a] | [L7-3a]           | [L7-3a] | [L7-3a]            | [L7-3a]     | [L7-3a] |

| $P(K)$ | Fixed   | Leave   | Classifier<br>(prior-corrected) | Oracle  | $R_{\text{const}}$<br>(best constant) | $R_Y^*$ | Classifier excess over $R_Y^*$ |
|--------|---------|---------|---------------------------------|---------|---------------------------------------|---------|--------------------------------|
| 0.20   | [L7-3a] | [L7-3a] | [L7-3a]                         | [L7-3a] | [L7-3a]                               | [L7-3a] | [L7-3a]                        |
| 0.30   | [L7-3a] | [L7-3a] | [L7-3a]                         | [L7-3a] | [L7-3a]                               | [L7-3a] | [L7-3a]                        |
| 0.50   | [L7-3a] | [L7-3a] | [L7-3a]                         | [L7-3a] | [L7-3a]                               | [L7-3a] | [L7-3a]                        |

Under confounding,  $R_Y^* = R_{\text{const}}$  exactly (Proposition 1), so the “classifier excess” column above is a direct empirical check of whether the prior-corrected classifier matches this floor — and it does, to within simulation noise at every prior (the one nonzero entry, at the highest prior, is sampling variance, not a systematic effect, and is not distinguishable from zero given the sample size used).

**Table 3. Expected cost per episode (toy units), SEPARATED regime.**

| $P(K)$ | Fixed   | Leave   | Classifier<br>(prior-corrected) | Oracle  | $R_{\text{const}}$<br>(best constant, <b>not</b> the achievable optimum here) | Classifier excess over $R_{\text{const}}$ |
|--------|---------|---------|---------------------------------|---------|-------------------------------------------------------------------------------|-------------------------------------------|
| 0.05   | [L7-3a] | [L7-3a] | [L7-3a]                         | [L7-3a] | [L7-3a]                                                                       | [L7-3a]                                   |
| 0.10   | [L7-3a] | [L7-3a] | [L7-3a]                         | [L7-3a] | [L7-3a]                                                                       | [L7-3a]                                   |
| 0.20   | [L7-3a] | [L7-3a] | [L7-3a]                         | [L7-3a] | [L7-3a]                                                                       | [L7-3a]                                   |
| 0.30   | [L7-3a] | [L7-3a] | [L7-3a]                         | [L7-3a] | [L7-3a]                                                                       | [L7-3a]                                   |
| 0.50   | [L7-3a] | [L7-3a] | [L7-3a]                         | [L7-3a] | [L7-3a]                                                                       | [L7-3a]                                   |

Here  $R_{\text{const}}$  is *not* the achievable optimum —  $Y$  is informative, so  $R_Y^*$  is

genuinely lower, and we have not computed its closed form for this simulated (non-Gaussian-summary) setting, only the analytical Gaussian benchmark in Section 4’s Corollary 2 discussion for the raw trajectory. The “classifier excess over  $R_{\text{const}}$ ” column shows the classifier substantially beats the constant-action baseline and sits close to the oracle, which is a legitimate and correctly-labeled finding — but it should not be read as “excess over the optimum,” since  $R_{\text{const}}$  is not the optimum in this regime.

Every “Fixed,” “Leave,” and “Oracle” value above matches the exact closed-form values  $pC_{OP} + (1 - p)\text{DISRUPT}$ ,  $p\text{HARM}$ , and  $pC_{OP}$  respectively, to full displayed precision — a direct, checkable correctness test of the cost computation.

### Findings:

1. **Neither constant policy is good away from its favorable regime.**  
Comparing Fixed against Leave shows an operator who always contains does badly when attacks are rare, and an operator who never contains does badly when they are common — a valid critique of any fixed, non-adaptive policy.
2. **Under separation, a properly prior-corrected classifier substantially beats the constant-action baseline and sits close to the oracle** — a finding about “excess over  $R_{\text{const}}$ ,” not “excess over the optimum.”
3. **Under confounding, our properly prior-corrected fitted classifier empirically matched the constant-action baseline, consistent with (but not logically guaranteed by) Proposition 1.**  
Proposition 1 establishes that the *Bayes-optimal* policy using  $Y$  equals the constant-action baseline under exact confounding — it does not guarantee that any given fitted policy will exactly attain this, since residual estimation or calibration error can in principle still produce a classifier strictly worse than the constant baseline even after correct prior adjustment. Our prior-corrected classifier approximately attained the floor in this experiment; we report this as an empirical finding, not as a restated theorem. A classifier that appears *actively worse* than the constant baseline under confounding is an artifact of applying a deployment-calibrated threshold to a training-prior-calibrated score, not a property of confounding itself. **The lesson is methodological:** cost-sensitive decision rules built on classifiers trained at one class balance and deployed at another must correct for that prior shift explicitly,

or they can appear to perform arbitrarily badly for reasons that have nothing to do with the underlying identifiability limit.

### 6.1. Sensitivity to the cost triple

Because  $(C_{OP}, \text{DISRUPT}, \text{HARM}) = (3, 6, 10)$  was chosen only to be strictly ordered and asymmetric, we re-ran the cost calculation of this section across seven triples spanning different ratios and overall scales — six genuinely additional triples plus the original  $(3, 6, 10)$  triple repeated here for direct comparison — at a fixed representative prior  $P(K) = 0.15$ , **using the prior-corrected threshold, not the deployment threshold applied to a training-prior-calibrated score.**

**Table 2b. Cost-of-ambiguity robustness across cost triples.**

| $C_{OP}$ | DISRUPT | HARM | Classifier<br>excess over<br>$R_{\text{const}}$ , CON-<br>FOUNDED | Classifier<br>excess over<br>$R_{\text{const}}$ , SEP-<br>ARATED |
|----------|---------|------|-------------------------------------------------------------------|------------------------------------------------------------------|
| 1        | 2       | 5    | [L7-3b]                                                           | [L7-3b]                                                          |
| 3        | 6       | 10   | [L7-3b]                                                           | [L7-3b]                                                          |
| 1        | 4       | 10   | [L7-3b]                                                           | [L7-3b]                                                          |
| 2        | 3       | 15   | [L7-3b]                                                           | [L7-3b]                                                          |
| 1        | 6       | 10   | [L7-3b]                                                           | [L7-3b]                                                          |
| 5        | 8       | 12   | [L7-3b]                                                           | [L7-3b]                                                          |
| 1        | 1.5     | 3    | [L7-3b]                                                           | [L7-3b]                                                          |

Across every triple tested, once the prior-correction is applied, **excess over  $R_{\text{const}}$  in the CONFOUNDED regime is zero or negligibly small (to within simulation noise) in every triple** — confirming, robustly, that Proposition 1’s guarantee holds for the constant-action floor and that no tested cost triple, not merely the original one, makes the prior-corrected classifier actively worse than a constant policy. **SEPARATED-regime excess is consistently and substantially negative across every triple** (classifier beats the constant-action baseline everywhere signal exists), confirming that finding’s robustness as well. As in Section 6,  $\text{CoA}^* = \min\{(1 - p)\text{DISRUPT}, p(\text{HARM} - C_{OP})\}$  is an exact, deterministic function of the cost triple and prior alone (no dependence on any classifier or data), included in Section 6’s discussion as context for checking the “excess” columns

against that floor, not as a quantity requiring its own empirical replication here.

## 6.2. An illustrative dollar-scale reconstruction (not an empirical calibration)

The cost constants used above are illustrative toy values. To give a sense of scale in real terms, we reconstruct a comparable triple from the Almgren–Chriss market-impact framework [27] (linear temporary-plus-permanent impact,  $L(X, T) = \epsilon X + \frac{\gamma}{2} X^2 + \eta X^2/T$ ), using the framework’s own original illustrative parameters (a \$50 stock, 5M-share average daily volume, \$0.125 spread, and the paper’s example impact coefficients) applied to three representative scenario sizes: a normal containment action (50,000 shares over 5 minutes), a needlessly disrupted benign order (the same size, forced into an urgent 5-minute unwind rather than a patient 3-hour one), and an unmitigated adversarial sequence (200,000 shares over a 30-minute exposure window). **We are explicit that this is an arithmetic reconstruction of an illustrative scenario using Almgren–Chriss’s own example parameters, not an empirical calibration to any measured trading incident, market, or venue.** In particular, the assumed containment trade (50,000 shares in 5 minutes) is a large share ([L7-3c]) of expected volume in that window under a uniform intraday-volume assumption — an aggressive execution schedule whose realism directly affects every downstream number, and which we have not validated against any specific real venue or instrument.

This reconstruction gives  $(C_{OP}, \text{DISRUPT}, \text{HARM}) \approx ([\text{L7-3c}], [\text{L7-3c}], [\text{L7-3c}])$  — satisfying the corrected Proposition 1’s requirement  $\text{HARM} > C_{OP}$ , but with one feature worth flagging explicitly:  **$C_{OP}$  and DISRUPT are nearly equal, with  $C_{OP}$  very slightly larger**, so the toy triple’s intuitive story (“needless disruption costs more than routine containment”) is not what this reconstruction shows; the disruption figure here mainly reflects a forced-urgency liquidation premium on a benign order, comparable to, not larger than, the containment action’s own cost.

**Applying the corrected Proposition 1 to this triple**, at  $P(K) = 0.05$ : the oracle’s expected cost is  $p C_{OP} \approx [\text{L7-3c}]$ , the best-constant baseline is  $\approx [\text{L7-3c}]$ , and the *optimal ambiguity cost* — the theorem-guaranteed floor,  $\min\{(1-p)\text{DISRUPT}, p(\text{HARM} - C_{OP})\}$  — is approximately  $[\text{L7-3c}]$ . We present this as an order-of-magnitude illustration only, not a measured or venue-representative figure, and any classifier’s *additional* regret beyond this floor must be measured with a properly prior-corrected policy (Section 6), not assumed equal to a naively-thresholded classifier’s total regret.

### 6.3. A corrected loss table for the audited fault mechanisms

Section 3.4 established that most of this paper’s simulated  $F$  mechanisms do not satisfy the “leave alone costs \$0” assumption their loss table requires — they submit real, market-affecting orders. We price this directly using the project’s Almgren-Chriss impact formula.

**Measured, not formula-predicted, volume.** The Almgren-Chriss impact coefficients  $(\eta, \gamma, \varepsilon)$  are legitimate, ADV/spread-derived constants for the stock in question; what matters is the *volume figure* fed into them, which must be the volume  $F$  actually submits within the 30-minute exposure window. A formula-predicted volume is unreliable here: in the zero-noise design (V2), where formula and ground truth can be compared with no measurement noise, the formula predicts several times the directly-logged 4,235 shares; in V1 it comes close to ground truth only by coincidence (two separate, opposite-signed errors nearly cancel for that design’s parameters); and in V3 and V4 its errors differ in sign and do not cancel.

**The costs below are therefore model-based costs using measured submitted volume, not a direct measurement of economic loss.** “Leave alone” costs are computed from measured simulated order volumes (the incident agent’s own logged submitted-order actions, not executed volume — cancellations and partial fills are not accounted for):  $H_F$  is the mean, over the 1000 fault episodes of each design  $\times$  regime cell of the paired basis, of each episode’s Almgren-Chriss cost of its directly-measured  $F$ -mechanism *submitted* volume (Table 1); these are the same episodes as the classification cells of Section 7.

Table 1: “Leave alone” cost for  $F$  from directly-measured submitted volume, per design and regime (paired basis, fault episodes of each cell): mean volume  $\bar{X}$  (sd),  $H_F$  with its 95% interval, the rebaselined fault cost of the sensitivity arm, and the cost-optimal threshold  $q^*$  under both conventions.

| Design              | Regime          | $\bar{X}$ (sd) | $H_F$ [95% CI]             | $C_F$ rebaselined [95% CI] | $q^*$ current | $q^*$ rebaselined |
|---------------------|-----------------|----------------|----------------------------|----------------------------|---------------|-------------------|
| V1_original         | contrast        | 30,972 (1,460) | 33,300 [33,105, 33,479]    | 189,523 [188,381, 190,565] | 0.348         | 0.159             |
| V1_original         | summary-matched | 55,653 (2,556) | 104,739 [104,146, 105,335] | 609,104 [605,609, 612,613] | 0.309         | 0.379             |
| Study4_expanded21ch | contrast        | 30,972 (1,460) | 33,300 [33,114, 33,488]    | 189,523 [188,438, 190,616] | 0.348         | 0.159             |
| Study4_expanded21ch | summary-matched | 55,653 (2,556) | 104,739 [104,133, 105,323] | 609,104 [605,528, 612,548] | 0.309         | 0.379             |
| V2_study5           | contrast        | 4,235 (0)      | 850 [850, 850]             | 3,764 [3,764, 3,764]       | 0.364         | 0.004             |
| V2_study5           | summary-matched | 4,235 (0)      | 850 [850, 850]             | 3,764 [3,764, 3,764]       | 0.364         | 0.004             |
| V3_third            | contrast        | 6,462 (531)    | 1,776 [1,760, 1,791]       | 8,607 [8,526, 8,692]       | 0.364         | 0.008             |
| V3_third            | summary-matched | 11,246 (701)   | 4,845 [4,811, 4,879]       | 25,477 [25,289, 25,666]    | 0.362         | 0.024             |
| V4_fourth           | contrast        | 12,497 (959)   | 5,906 [5,856, 5,964]       | 31,435 [31,153, 31,757]    | 0.362         | 0.030             |
| V4_fourth           | summary-matched | 34,354 (2,610) | 40,874 [40,513, 41,257]    | 233,763 [231,653, 236,007] | 0.344         | 0.189             |

**Threshold versus realized cost.**  $q^*$  itself is a property of the cost structure alone, independent of the deployment prior (under the current

convention it stays between 0.309 and 0.364 across every design and regime; Table 1); this concerns the threshold formula, not how often the *classifier’s realized cost* falls below a constant policy. Section 7.7 reports how often the classifier’s realized cost falls below each constant policy, with intervals.

**What this table computes.** The Almgren-Chriss impact function used throughout this paper includes both temporary and permanent impact at fixed duration  $T$ :  $L(X, T) = \varepsilon X + \left(\frac{\gamma}{2} + \frac{\eta}{T}\right) X^2$ . By Jensen’s inequality, since this is convex in  $X$ :  $\mathbb{E}[L(X, T)] = L(\mathbb{E}[X], T) + \left(\frac{\gamma}{2} + \frac{\eta}{T}\right) \text{Var}(X)$ . Pricing the *mean* measured volume is not the same as the *mean* of the per-episode costs, and understates the latter whenever volume varies episode-to-episode (the deterministic V2 design is exempt from this specific gap, since its volume has zero variance by construction).

**The  $H_F$  of Table 1 is the mean of per-episode costs, not the cost of the mean volume:** each fault episode’s volume is priced with the full impact function, linear term  $\varepsilon X$  included, and the costs are averaged, so the variance term above is included by construction. It is nonzero for every stochastic design and zero for the deterministic V2 design, whose volume has zero variance (Table 1). These are the  $H_F$  values used by every cost table of Section 7.7.

The table’s  $H_F$  should be read as “the mean cost of executing the measured volumes, at fixed duration  $T$ ,” which is a well-defined and useful quantity in its own right, not as a claim to have captured every source of episode-level cost variation; for variable duration the expectation involves  $X^2/T$  and does not reduce to this fixed- $T$  identity without further assumptions.

**A plausible-looking formula-derived number should be checked against direct measurement before being trusted, not merely checked for internal consistency;** the figures in this section meet that standard.

**Consequence for Section 3.4’s audit:**  $F$ ’s submitted order volume is not literally free to price at \$0 under this model — the measured-volume-based costs (from \$850 to \$104,739, Table 1) are still nonzero and still larger than the original loss table’s assumed \$0 — so the loss table in Sections 6–7 remains an illustrative simplification, not a validated economic costing, of what these specific simulated  $F$  mechanisms actually do or the incremental harm they cause relative to a baseline. **Consequence for the policy comparison:** per the full 50-cell audit (Section 7.7, using the direct class-conditional weighting method), the classifier has a lower point-estimate cost than always-contain in all 50 cells (a supported improvement in 49) and is

never worse than the best constant policy with a supported interval; its advantage over the best constant policy is a supported improvement in 40 cells, and the cells without one lie at the low end of the prior grid or in the Study 5 profile (Section 7.7).

## 7. An empirical study of observation matching in a realistic non-linear market

**Reproducibility status of the results in this section.** An RNG audit (Section 7.10.5) found and patched **further unseeded-RNG bugs** beyond ABIDES’s `get_wake_time()` issue; bit-identical cross-process reproducibility holds only *after* all of them are fixed. **A further failure mode: worker processes cache their first lazy import of the patched code for their entire lifetime, so results generated after the on-disk patch could still silently reflect pre-patch behavior depending on a worker’s exact import timing — a failure mode invisible to file-timestamp-based provenance checks.** It affected the frozen confirmation set (Section 7.10.3), which is therefore **not provenance-clean** and not used. A hardened pipeline (spawn, one task per worker, forced fresh import, and a runtime behavioral fingerprint confirming all the RNG patches are active, independent of timestamps) was then built. **The ten headline classification cells (Table 2) and the cost table built on them are generated with this pipeline on paired seeds (Section 7.10.5). The other ABIDES results of Sections 7.1–7.8, and the severity and dose experiments of Section 7.10, predate it:** they should be understood as statistically valid (genuine, if not fully deterministic, randomness) but not bit-for-bit reproducible, and are reported as exploratory.

**What this section does and does not test.** Theorem 1 concerns equality of the *entire* observation trajectory law,  $P_K(Y_{0:T}) = P_F(Y_{0:T})$ . What the constructions below actually do is match the two mechanisms on *one or a few chosen summary statistics* (mean price drift, order count, mean spread) while other observable channels (order counts, depth, cancellation rate, size) remain measurably different. **Matching one moment is not equality of the joint trajectory distribution, and a classifier exploiting an unmatched channel does not contradict Theorem 1 — this would be true even in a fully linear system.** A trivial counterexample makes this concrete:  $Y \mid K \sim \mathcal{N}((0, 2), I)$ ,  $Y \mid F \sim \mathcal{N}((0, -2), I)$  have identical first-coordinate means but a highly informative second coordinate; no nonlinearity

is needed for this pattern.

Accordingly, we call the constructions below **summary-matched** (matched on one or more chosen statistics; labelled CONFOUNDED in the tables below) and **contrast** (deliberately given a large chosen-attribute difference; labelled SEPARATED), and we do not claim any comparison between them is a test of Theorem 1’s prediction. What this section actually establishes — and this is still a real, useful finding — is an empirical account of **which observable channels leak information under which summary-matching choices**, and how sensitive that leakage is to feature selection, calibration target, classifier choice, and incident footprint. Section 7.3’s synthesis and Section 7.4’s practitioner checklist are stated in these terms. We also run an ABIDES-based **negative control**: two mechanisms with *identical* observable order-generation kernels (not merely matched on one statistic) should show no repeatable discriminatory signal under valid, independently-generated evaluation data; a persistent signal there would indicate an implementation or evaluation-boundary problem. The identical-kernel negative control (Section 7.10.1) shows no detectable accuracy signal at the pre-specified margin; its AUC interval does not meet that margin. It checks the pipeline and evaluation boundary, not Theorem 1.

The linear model’s Theorem 1 remains exact and, per Sections 5–6, reproducible within its own stated scope. This section does not test whether Theorem 1 is true — it is proved, not tested — nor does it establish a validated “linearization” bridge from Theorem 1’s model to ABIDES’s dynamics (no operating point, Jacobian, or approximation-error bound connecting the two has been derived; that would be required to make a linearization claim, and we have not made one). What we did was inject compromise/fault mechanisms as real order attributes (submission rate, order size, price aggressiveness, directional bias, cancel/replace churn) into ABIDES [28], a real discrete-event multi-agent market simulator, and study the resulting classification and cost behavior as an object of independent empirical interest — a comparison of a stylized linear model and an agent-based one, not a validation or falsification of the former by the latter.

*7.1. First result: summary-matching on one statistic leaves other channels informative*

**Matching mean price drift alone does not come close to matching other observables.** At the calibrated magnitude, the two mechanisms’ mean order counts differed substantially, cascading into large gaps in depth

and spread statistics the calibration search did not control. Two independent calibration searches, differing only in random seed, selected different “best” magnitudes for even this one matched statistic — the discovery process itself is noisy.

**A passive classifier discriminated the summary-matched scenario more accurately than the contrast scenario.** On the paired Era-3 basis (Table 2; 1000 episodes per class, 600 held-out test episodes from 300 test seeds), accuracy is 0.857 (interval 0.830–0.882; AUC 0.934) on the summary-matched construction, versus 0.685 (interval 0.652–0.718; AUC 0.756) on the contrast construction — the opposite ranking from the pilot’s Table 1 pattern, and worth investigating for that reason, but **not, on its own, evidence against Theorem 1**, since the summary-matched construction was never shown to satisfy Theorem 1’s actual premise. The two accuracy intervals do not overlap. Earlier, smaller-sample regenerations of the contrast cell, generated before the hardened pipeline, varied widely, one of them reading below chance; they have no retained sample size or interval, are not used as evidence, and their spread is not diagnosed here.

The cost-of-ambiguity quantification (with the prior-shift correction of Section 6) inherits the same ranking; this too should be read as “the summary-matched scenario as constructed left more decision-relevant information available to this specific policy than the contrast scenario did” — not as a violation of any theorem.

Table 2: Classification on the paired Era-3 basis: accuracy and AUC for the five profiles in both regimes. The ten classification cells are computed on paired seeds: each seed yields one compromise and one fault episode, the train/test split is grouped by seed, and the intervals are cluster-bootstrap intervals over seeds, all under the hardened pipeline (Section 7.10.5). <sup>†</sup> Zero test errors: exact Clopper–Pearson interval over the test episodes (the bootstrap interval is degenerate; Appendix A).

| Design              | Contrast             |                      | Summary-matched                   |                      |
|---------------------|----------------------|----------------------|-----------------------------------|----------------------|
|                     | Accuracy [95% CI]    | AUC [95% CI]         | Accuracy [95% CI]                 | AUC [95% CI]         |
| V1_original         | 0.685 [0.652, 0.718] | 0.756 [0.728, 0.786] | 0.857 [0.830, 0.882]              | 0.934 [0.917, 0.950] |
| Study4_expanded21ch | 0.998 [0.995, 1.000] | 1.000 [1.000, 1.000] | 0.898 [0.875, 0.920]              | 0.966 [0.954, 0.977] |
| V2_study5           | 0.663 [0.633, 0.692] | 0.731 [0.706, 0.757] | 0.553 [0.525, 0.580]              | 0.577 [0.551, 0.604] |
| V3_third            | 0.998 [0.995, 1.000] | 1.000 [1.000, 1.000] | 0.875 [0.850, 0.898]              | 0.953 [0.940, 0.967] |
| V4_fourth           | 0.727 [0.698, 0.757] | 0.807 [0.783, 0.832] | 1.000 [0.994, 1.000] <sup>†</sup> | 1.000 [1.000, 1.000] |

This first result raised an obvious question: is this ranking a robust property of these two specific constructions, or an artifact of *this particular* implementation’s choices? Section 7.2 answers this with a systematic sensitivity

analysis, deliberately designed so that no single run could be mistaken for the final word.

## 7.2. Sensitivity analysis: five extension studies

**Study 1 — Statistical robustness of the accuracy ranking.** In an earlier, pre-hardening replication over five independent seeds, bootstrap intervals on the accuracy and AUC gaps (computed over the five seeds’ held-out sets) both excluded zero, in the same direction as the paired-basis cells of Table 2, whose accuracy intervals do not overlap. The cost-metric version of this ranking was directionally consistent but noisier (a few seed  $\times$  prior combinations flip sign at extreme priors) — a seed-level comparison that predates the prior-shift correction of Section 6 and has not been recomputed with it. The corrected ABIDES cost analysis, with the classifier-excess-over-best-constant decomposition, is Section 7.7’s, on one evaluation split per profile rather than on these seeds.

**Study 2 — Classifier-family sensitivity.** The ranking holds for three of four classifier families: logistic regression, Gaussian naive Bayes, and random forest. The fourth, a naive single-feature threshold rule, ranks the two scenarios the other way when properly oriented, a mild dissent from the other three classifiers’ ranking; with a hardcoded direction its accuracy falls below chance, the signature of an inverted decision boundary. We read this as evidence that the ranking’s strength depends on classifier capacity to exploit whatever unmatched channels are present, not as evidence about any theorem.

**Study 3 — Population and scale sensitivity.** The ranking holds across the range of background population sizes tested and at double session length, ruling out fragility to one arbitrary population scale.

**Study 4 — Feature-set sensitivity (the pivotal result for this section).** Adding order-flow imbalance [29], signed executed volume, and queue concentration **reverses the ranking again**: on the paired basis the contrast scenario’s accuracy reaches 0.998 (interval 0.995–1.000, 1 of 600 test episodes misclassified), while the summary-matched scenario sits at 0.898 (interval 0.875–0.920; Table 2). **Correct interpretation:** the contrast scenario was, in fact, always highly distinguishable on the attributes it was designed to differ on — the original 4-channel feature set (a *channel* is one observed per-time-bin series: spread, depth, cancellation rate and mid-price in the original set; a classifier’s *features* are the mean, standard deviation and

net drift of each channel over the episode) simply failed to expose that difference, while an unrelated activity-level mismatch in the summary-matched scenario happened to leak through those same 4 channels more strongly. Adding the right features corrects the *feature-access* problem, not a property of either scenario’s underlying observation law. One data-availability note: ABIDES logs per-trade execution prices (a dedicated check found them in every tested trade bin), so a literal volume-weighted-average-price (VWAP) feature is available; no VWAP-feature comparison is reported in this paper. The signed-volume proxy used in this study is a reasonable feature choice on its own merits, independent of any VWAP comparison.

**Study 5 — A second injection design.** Built from the attribute specification alone, using a different attribute subset, a deterministic wake-up clock, and a different calibration target. **Chance, as used from here on:** a cell is “at chance” when its accuracy interval contains one half, and “above chance” when the interval lies above it. On the paired basis (Table 2) both Study 5 cells are above chance, but weakly: the contrast case reaches 0.663 accuracy (interval 0.633–0.692; AUC 0.731) and the summary-matched case 0.553 (interval 0.525–0.580; AUC 0.577), the weakest of the ten cells. An earlier five-seed replication, generated before the hardened pipeline, had read the summary-matched case as at chance and the contrast case as weakly above it. **A caution applies to any such reading:** an interval containing one half means the tested procedure did not detect a reliable departure from chance — it does not establish that the two mechanisms’ observation laws are equal, and a wide interval leaves room for a practically meaningful amount of undetected signal. Earlier smaller-sample estimates of the contrast case were lower than the paired-basis value; because they differ from it in both sample size and pipeline, the difference is not attributed to either.

**Terminology note for Sections 7.3–7.10 below:** the labels **CONFOUNDED** and **SEPARATED** appear throughout the remaining subsections as inherited shorthand for, respectively, the summary-matched and contrast constructions defined above. Neither label should be read as a claim that Theorem 1’s premise (full trajectory-distribution equality) was established or tested for either construction — they denote specific engineering choices (which statistic was matched, which was left free to differ).

*7.3. Synthesis: what these constructions show about observation-matching, not about Theorem 1*

In Section 7.2, three genuinely different classification profiles emerged from three reasonable implementation choices — the original construction (accuracy ranking one direction, confirmed with high statistical confidence), a richer-feature version of the same construction (ranking reversed, also confirmed with high statistical confidence), and a second injection design (weak discrimination in both scenarios, weakest in the summary-matched one). Two further injection designs (Sections 7.5 and 7.9) each add a distinct profile, for five classification profiles across four injection designs, all parameterizations of one shared simulation infrastructure (Section 9) (Study 4 shares the original design’s mechanism and differs only in feature access). The threshold-rule baseline in Study 2 adds that even *within* one fixed construction, a weaker classifier can land on a different ranking than stronger ones do on the same data. We do not treat any single one of these as *the* answer; the pattern across them — that the ranking flips depending on engineering choices with no analog in the linear model — is the finding.

**The mechanism is structural, and it concerns observation-matching difficulty, not theorem validity.** In the linear model, one vector equality (in  $\mathbb{R}^m$ , the observation dimension — not literally “one scalar equality” unless the operator has only a single observable channel) zeroes out the difference between mechanisms in *every* observed coordinate at *every* time step simultaneously, because  $C$  is fixed, known, and identical across time, and — per Section 4.1’s Theorem 2 — that equality is itself feedback-fragile except under non-generic dynamics. In a nonlinear multi-agent market, there is no single fixed linear map to match in the first place: an engineer attempting to construct an analog of exact confounding must choose a finite set of statistics to match (mean drift, order count, spread), and whatever is left unmatched can leak. Section 7.2’s Studies 4 and 5 (with confidence intervals, not single-run estimates) show this leakage depends on at least two further choices — **which, per a direct audit, are representable but unexplored in this paper’s linear-model pilot, not structurally absent from the linear model: the richness of the observer’s feature set** has a direct linear-model analog in the number of rows of  $C$  (how many gateway-observable channels the operator has), which was never expanded beyond its original two rows in any linear-model script in this project, even though ABIDES’s analogous feature-richness axis (Study 4) was explicitly swept; and **the incident’s footprint size relative to the background**

**population** has a functional analog in the ratio of injection magnitude to the process/observation noise scale ( $Q, \Sigma$ ), which the linear-model pilot swept the numerator of (Table 1’s magnitude sweep) but never the denominator of — the noise scale was hardcoded and identical across every linear-model script in this project. **A third factor, from Study 2, is classifier capacity**, similarly explored on both sides: Study 2 (Section 7.2) tested four classifier families on ABIDES data (logistic regression, Gaussian naive Bayes, random forest, and a threshold rule) — the same count as the linear-model pilot’s Table 1b. **The one factor genuinely absent from the linear model, not merely unexplored, is the calibration-by-search mechanism ABIDES’s summary-matched construction needs** — the linear model’s exact confounded case is constructed analytically via  $C$ ’s nullspace, with no search required, because linearity admits a closed-form solution; ABIDES’s nonlinear mapping from attributes to summary statistics has no such closed form, so its construction genuinely requires an approximate grid search the linear model structurally does not need. We have not disentangled how much of this instability traces to Theorem 2’s state-feedback mechanism specifically versus these factors — Section 7.8’s ablation attempted this and remains inconclusive (Section 7.10.2) — and separating their relative contributions is a natural next step, not a claim we make here.

**What the evidence actually supports.** Across every tested construction, genuine chance-level discrimination (consistent with Theorem 1’s confounding prediction) appeared in one case: the exploratory one-off ablation (Section 7.8), where reduced persistence, duration, and cumulative activity together — not state-feedback alone (Section 7.8) — brought the summary-matched case down to a level consistent with chance. **We do not attribute Study 5’s summary-matched behavior to footprint size: Section 7.5’s dedicated sweep found footprint alone does not explain it, and we have no clean isolated explanation for it.** This is a statement about **how difficult it is to construct a valid nonlinear instantiation of exact confounding**, not a statement that Theorem 1 fails once instantiated — since none of the summary-matched constructions were shown to be valid instantiations in the first place. The claim is: matching a small number of chosen statistics between two mechanisms in a realistic market is not a reliable way to produce the kind of full observational equivalence Theorem 1 describes, and attempting it is easy to get wrong in ways that are hard to predict from the linear model alone.

**What this means going forward.** Theorem 1 remains an exact, useful

diagnostic for the linear-Gaussian model it is proved for; nothing in Section 7 bears on that. What Section 7 does establish is a caution for anyone wanting to build a nonlinear analog of a linear identifiability result for validation purposes: doing so correctly requires an explicit account of which statistics are matched, which features an observer can access, and how much of the incident’s aggregate footprint remains after matching — none of which the linear model specifies, and none of which we found a way to get right by intuition alone across our four *distinct-mechanism* attempts (the original, Study 5, and the third and fourth designs, all of which only ever matched a chosen summary statistic between genuinely different mechanisms). **One narrower construction satisfies full observation-law equality by construction:** Section 7.10.1’s negative control uses two labels with *identical* observable order-generation kernels, not merely a matched summary statistic; its result is stated at the opening of Section 7 and reported in Section 7.10.1. The open, unresolved question is narrower than “whether any valid instantiation exists at all”: it is whether a nonlinear construction with two genuinely *different* underlying mechanisms (not merely differently-labeled identical behavior) can be engineered to validly satisfy Theorem 1’s premise — that is the question our four distinct-mechanism attempts did not resolve, not the question of whether identical-kernel equivalence is achievable in a nonlinear simulator at all, which holds trivially by construction.

#### 7.4. A practitioner checklist

Grounded strictly in what the five extension studies of Section 7.2, the second-round experiments of Section 7.5 and their replications showed, not extrapolated beyond the data:

1. **Do not assume a linearized identifiability analysis’s premise is easy to instantiate correctly in a nonlinear deployment.** Across every attempted construction in this study, none was confirmed to actually satisfy Theorem 1’s premise (full observation-law equality) — engineering that equivalence correctly, rather than merely matching one or two convenient statistics, is itself a nontrivial and easy-to-get-wrong task.
2. **Feature richness matters, but the specific feature matters far more than richness per se.** Section 7.5 isolates signed executed volume as the single feature responsible for Study 4’s entire ranking

recovery; order-flow imbalance and queue concentration, added individually, recovered nothing. Don't chase "more features" — chase the specific microstructure quantity your feature set is missing.

3. **Footprint size, on its own, is not the deciding factor it first appeared to be.** Section 7.5's dedicated sweep found the reversal present across the range of incident footprints tested using the original feature set — footprint alone does not explain Study 5's different behavior. Don't assume a small-footprint incident is automatically the hard case; check the feature set first.
4. **Don't rely on a single classifier as a safe default without checking a stronger one alongside it.** Study 2 shows a weak classifier can land on a different ranking than a stronger one on the *same* data — a single baseline's result is not conclusive evidence about the underlying identifiability structure.
5. **Replicate across seeds before trusting any single run's ranking.** The original Study 5 single-run read as "no signal anywhere"; replication across seeds, and later the paired basis, revealed a real, if weak, signal. A single calibration/simulation draw is not sufficient evidence either way.
6. **Population size and session length are lower priority to check than feature set.** Study 3 found the reversal stable across the population range tested, and Section 7.5 found it stable across a wide footprint range too — feature choice, not scale, is where the real sensitivity lives.
7. **There is no theory-only way to know which regime you are actually in, and the dichotomy itself may not be stable.** Four injection designs, all parameterizations of one shared infrastructure (the original, Study 5, Section 7.5's third design and Section 7.9's fourth design), plus one feature-engineering variant of the original (Study 4), produced five qualitatively different classification profiles. This is not merely "hard to predict in advance" — it suggests the confounded/separated labels do not have one fixed operational meaning across reasonable real-market constructions. Section 7.6 gives the rationale for this set of designs.
8. **Investigate surprising results before either dismissing or trusting them — and note that "uninformative" is not the same as "low stakes."** Section 7.5's below-chance footprint result was checked with the same rigor as the Study 2 threshold-rule inversion and held at the lowest footprint level tested under independent replication, but

not at a nearby, moderately higher footprint level (Section 7.5). More importantly, Section 7.7’s cost extension shows the weakest classifier (Study 5) is not a moderate-cost middle ground — it holds most of the cells where the classifier adds no supported value over the best constant policy (Section 7.7 gives the per-cell comparison). Don’t assume every surprising number is an error, don’t assume every surprising number is real, don’t assume ten resplits of one dataset are ten independent checks, and don’t assume a “weak signal” result is a low-stakes one — check with genuinely independent data, and cost it out.

9. **Verify your simulator’s determinism claims independently of your own code.** Section 7.5 found a real reproducibility bug in ABIDES’s own RNG handling (`get_wake_time()` uses an unseeded global generator) that breaks bit-for-bit reproducibility across process runs despite this project’s own code seeding correctly throughout — “my code seeds everything” is not the same claim as “my results reproduce exactly.”
10. **Check whether your model injects the incident at the output or the state level before trusting any confounding claim.** Section 4.1’s Theorem 2 shows this distinction alone can flip an exactly-confounded pair into a highly classifiable one, with no nonlinearity required. The operative distinction is *state-level vs. output-level* injection, not *persistent vs. one-time* — a one-time state-level perturbation is governed by the same unobservability condition as a persistent one (Section 4.1, qualification 1). A state-level disturbance, whether one-time or repeated, is generically far more fragile to confound than an output-level one, and this should be checked before assuming a theoretical confounding result transfers even to a linear operational setting, let alone a nonlinear one.

#### 7.5. *A second round of validation: pinpointing the drivers, and one new reproducibility finding*

A second, more targeted round of experiments converts the qualitative claims above into precise, checked ones, and sharpens two of them.

**Feature richness: one specific feature, not general richness, drives Study 4’s recovery.** Testing each of Study 4’s three added features individually against the original 4-channel baseline isolates the effect precisely: adding *signed executed volume alone* reproduces the full ranking recovery

(SEPARATED near-perfect, above CONFOUNDED), while adding order-flow imbalance alone or queue concentration alone recovers nothing (both stay at baseline, SEPARATED still below CONFOUNDED). This sharpens Section 7.3’s framing: what matters is whether the observer’s feature set captures signed, executed (not merely resting) order flow — a specific microstructure quantity, not feature-set size in general.

**Footprint size does not explain the reversal — it was the wrong hypothesis.** A dedicated sweep, varying the incident’s aggregate order-flow footprint from a small to a large share of total market volume using the *original* 4-channel feature set, found the reversal (CONFOUNDED > SEPARATED) present at essentially every footprint level tested, at small and large footprints alike. Footprint alone, holding the feature set at baseline, does not turn a reversal into a recovery or vice versa. Whatever distinguished Study 5’s design must trace to some other construction choice (attribute subset, calibration target, or clock structure); we do not have a clean isolated explanation for it beyond what Section 7.2 already reported.

At the very lowest footprint tested, SEPARATED accuracy fell *below* chance. Investigated with the same scrutiny applied to the Study 2 threshold-rule bug: this is **not** a hardcoded-sign artifact (this classifier learns its orientation from training data). Resampling further train/test splits of the *same* dataset (most of them below chance) is not an adequate replication: resplits of one fixed dataset are not independent experiments, since they share the same underlying episode pool, and at the next footprint level that method suggests the same. A check using genuinely independent data realizations at the two lowest footprint levels finds the pattern does not hold uniformly — most independent replicates fall below chance at the lowest level, but at the next level they split in a way consistent with chance. **We read the lowest-footprint result as more likely genuine, but do not treat a broader “consistently below chance across low footprints” pattern as established.**

**A fourth, distinct outcome from a third injection design.** A third incident agent (built from the attribute specification only, varying a different attribute subset and calibration target than either the original design or Study 5) produced yet another distinct pattern: SEPARATED reached near-perfect accuracy using only the *original* 4-channel feature set, while CONFOUNDED simultaneously reached a high (though not near-perfect) level, although its calibration target (total order count) was not matched more closely than the other designs’ targets: on the common tolerance audit

of Section 7.6 (Cohen’s  $d$  on each design’s own target statistic, independent data), it is farther from its target than the original design and Study 5.

**On the paired basis (Table 2) the separated regime reaches 0.998 accuracy (AUC 1.000) and the confounded regime 0.875 (interval 0.850–0.898; AUC 0.953).** This design’s configuration builder had not been patched for the RNG bugs before the hardened regeneration (Section 7.10.9), and the two regimes responded differently to the patch: the separated regime was essentially unchanged, **ruling out the RNG bug as an explanation for this design’s near-perfect separated-regime performance**, while the confounded regime fell, meaning **part of this design’s pre-patch confounded-regime near-separability was an artifact of the unpatched RNG, not a property of the construction**. The qualitative finding (this design discriminates well in *both* regimes, unlike the others) holds on the paired basis.

The mechanism is directly diagnosable: this design’s compromise and fault mechanisms differ sharply in `cancel_rate`, which is directly one of the four observed channels, so matching total order count left cancellation behavior free to leak a strong signal. **The first three injection designs (the original, Study 5, and this design), together with one feature-engineering variant of the original (Study 4), produce four qualitatively different classification profiles:** reversed (original, baseline features), fully recoverable via one specific feature (original design, expanded features — Study 4), a muted middle ground (Study 5), and simultaneously strong-on-both-regimes via a different leaking channel (this third design); the fourth design (Section 7.9) adds a fifth, for five profiles across four designs. We read this as direct evidence that the confounded/separated dichotomy does not have one fixed operational meaning in this class of nonlinear market — which channel leaks, and how much, depends on exactly which attributes a designer chooses to vary versus match, in a way the linear model’s fixed, given  $(C, b_K, b_F)$  never has to confront.

**A separate reproducibility finding, unrelated to the substantive results.** While investigating the below-chance footprint result, attempting to regenerate it revealed that identical seeds do not produce bit-identical episodes across separate process invocations of this codebase, traced to ABIDES’s own `get_wake_time()` utility (scheduling most of the background population’s wake-up times) calling Python’s unseeded global random number generator rather than the codebase’s own correctly-seeded random state objects. This is a bug in ABIDES’s third-party code, not in any

script written for this project. **This bug, and two further unseeded-RNG bugs, are patched (Section 7.10.5)** — see that section and the Appendix for the reproducibility status; this specific below-chance result was generated before the fix, so this particular result and everything in Sections 7.1–7.8 should be read with that caveat, distinct from the `str_to_ns` bug (Appendix), which blocked correctness outright and had to be fixed before any ABIDES results could be trusted at all, versus the RNG bugs, which affect only bit-for-bit reproducibility of specific seeds, not the validity of any individual run’s statistics.

**A note on the broader recognition of this transfer question.** Section 2 discusses the Lie-group transfer condition of Alisic & Amin [17]. The point relevant to this section is that whether a property established under one operating condition carries over to another is a live concern in the adjacent literature, not an artifact of this paper’s particular framing.

#### *7.6. Why these injection designs: a stopping-rule justification*

The first three injection designs were a deliberate stopping point, not an arbitrary one, and not merely a budget constraint. The justification: the opening of Section 7 names five controllable dimensions of an ABIDES incident (directional bias, size, rate, price aggressiveness, cancellation churn) — these are the concrete, simulator-specific instantiations of the abstract injection direction  $b_M$  in Section 3.2’s linear model, not a table given in Section 3.2 itself, which states the general model only. The three designs tested differ in exactly the way that matters for this paper’s argument — which subset of those five dimensions is varied versus held fixed during calibration, and against which target statistic calibration is performed. The original design varies all five and calibrates against mean price drift; Study 5 varies a different subset with a different (spread-fraction, deterministic-clock) parameterization and a different calibration target; the third design varies yet another subset and leaves `cancel_rate` uncalibrated, which is precisely what produces its distinctive leakage. Between them, the three designs span the qualitatively distinct calibration failure modes identified across this section: leakage through activity-level mismatch (original), an as-yet-unexplained muted signal (Study 5 — **not explained by footprint size**, per Section 7.5’s dedicated sweep), and leakage through an uncalibrated discrete-behavior channel (third design). We do not claim these three exhaust the design space: **a fourth design, built specifically to test this, surfaced a fifth profile (Section 7.9)**, so the evidence base is four designs

and five profiles. What we do claim is that the first three independently-motivated designs, chosen to differ along the five dimensions named at the opening of Section 7 as the relevant attack surface, already established the paper’s core claim (the dichotomy is construction-dependent, not fixed) with enough distinct failure modes to rule out “we got unlucky once” as an explanation, and that the fourth design is consistent with it — a claim about instability, not about a complete taxonomy of that instability, which remains future work.

**A matching-tolerance audit sharpens what “matched” means here and should be read alongside the above.** No pre-specified numerical tolerance was set before this project began measuring how close each design’s calibration search actually got to its target statistic. Auditing all four designs, on independent (non-calibration) data, against even a generous tolerance ( $|Cohen’s\ d| < 0.1$  on the target statistic): **zero of four designs achieve it**; the original design is closest, followed by Study 5, then the third design (target: total order count), and the fourth design is farthest. These values are Cohen’s  $d$  on each design’s own target statistic in independent (non-calibration) data; they are not the pooled-standard-deviation gaps on the calibration data discussed in Section 7.9. **Every instance of “matched” or “summary-matched” in this section should be read as “match attempted, not match achieved to any pre-specified standard”** — not a successful match followed by unexpected leakage. **For the closest design (V1), a finer, 16-point calibration grid searched against independent held-out data at every point gives an apparent match at  $\theta = 1.8$ , but that value was *selected* as the best-looking candidate among 16 grid points, so it is not an untouched confirmatory result, even though the grid’s underlying episodes were independently generated.** Fresh confirmation runs at this *same selected*  $\theta = 1.8$  (not re-running the selection) do not reproduce it: a first re-run, a second, larger ( $N = 150/\text{class}$ ) one, and ten further independent fresh batches ( $N = 150/\text{class}$  each) give twelve independent  $d$ -estimates **that are all negative and tightly clustered, far from the grid search’s single apparent “hit”**; a one-sample  $t$ -test rejects a mean of zero by a wide margin ([L7-13]). This is a systematic, statistically robust non-zero effect, not noise around zero:  $\theta = 1.8$  consistently overshoots, making  $F$ ’s mean drift substantially exceed  $K$ ’s — the opposite of the intended match — and the original grid search’s apparent success is now conclusively a small-sample fluke, not a real calibration point. This does not establish that no fixed  $\theta$  can

meet tolerance for this design, or that the result generalizes to the other three designs, but it does close the specific question of whether  $\theta = 1.8$ 's failure was noise (it is not) — and we did not pursue the same exercise for the other three (already worse-matched) designs. The settled claim — the one selected calibration point for the closest design fails tolerance systematically, not by chance — is consistent with the paper's core argument that engineering a valid nonlinear analog of exact confounding is difficult.

*7.7. Cost tables with positive fault loss (illustrative dollar-scale reconstruction, not an empirical calibration)*

Table 3 applies the positive fault loss of Section 6.3; Section 6.2's caveat applies identically here. It uses the illustrative dollar cost triple of Section 6.2 ( $C_K = C_{OP}$ ,  $C_F = \text{DISRUPT}$ ,  $H_K = \text{HARM}$ ) together with the directly-measured, design-and-regime-specific  $H_F$  values of Table 1, for five profiles (four injection designs plus Study 4, the feature-engineering variant of the original design)  $\times$  2 regimes  $\times$  5 priors (50 cells). **The best constant policy's excess over the oracle agrees with the exact accounting identity  $B(p) = \min\{p\Delta_K, (1-p)\Delta_F\}$ , with  $\Delta_K = H_K - C_K$  and  $\Delta_F = C_F - H_F$ , to floating-point precision in every cell.**

Study 4 shares its underlying incident mechanism bit-for-bit with the original design (V1) — `action_log` output under identical seed, mechanism, direction and magnitude matches exactly, including total order count and shares — so it correctly reuses V1's ground-truth-measured  $H_F$  rather than requiring independent measurement; only its feature set (the original 4 channels plus the three channels added in Study 4) differs.

**Cost estimator: direct class-conditional weighting**,  $\hat{R}_p = p\{C_K + (H_K - C_K)\hat{\text{FNR}}\} + (1-p)\{H_F + (C_F - H_F)\hat{\text{FPR}}\}$ , **computed once per cell from the held-out test set's FPR/FNR at the fixed cost-optimal threshold — with no random resampling step, so no resampling-draw noise.** Intervals on the differences between policies are cluster-bootstrap intervals over test seeds, the two episodes of a seed moving together; the constant policies' costs are not resampled.

**Study 4's richer feature set produces a larger cost advantage than the original 4-channel set in the contrast regime** (Table 3: every Study 4 contrast cell is a supported improvement over the best constant policy, while the original design's two lowest-prior contrast cells are inconclusive), consistent with Section 7.2's finding that the expanded feature set

Table 3: Full cost table on the paired basis (current fault-cost convention), one row per design  $\times$  regime  $\times$  prior: the classifier’s cost  $R_{\text{clf}}$ , the best constant policy’s cost, their difference with a cluster-bootstrap 95% interval over seeds, and the interval verdicts against the best constant and against always-contain. Costs in dollars (illustrative reconstruction, Section 6.2).

| Design              | Regime          | Prior | $R_{\text{clf}}$ | Best constant     | Difference [95% CI]           | vs best constant | vs always-contain |
|---------------------|-----------------|-------|------------------|-------------------|-------------------------------|------------------|-------------------|
| V1_original         | contrast        | 0.05  | 99,906           | 97,510 (leave)    | 2,395 [−1,791, 7,564]         | inconclusive     | improvement       |
| V1_original         | contrast        | 0.10  | 160,009          | 161,720 (leave)   | −1,711 [−7,827, 5,451]        | inconclusive     | improvement       |
| V1_original         | contrast        | 0.20  | 264,300          | 290,140 (leave)   | −25,840 [−39,246, −12,427]    | improvement      | improvement       |
| V1_original         | contrast        | 0.30  | 341,975          | 418,560 (leave)   | −76,585 [−94,347, −59,651]    | improvement      | improvement       |
| V1_original         | contrast        | 0.50  | 429,586          | 482,448 (contain) | −52,861 [−68,020, −36,697]    | improvement      | improvement       |
| V1_original         | summary-matched | 0.05  | 145,395          | 165,377 (leave)   | −19,982 [−24,318, −14,687]    | improvement      | improvement       |
| V1_original         | summary-matched | 0.10  | 182,556          | 226,015 (leave)   | −43,460 [−51,202, −34,874]    | improvement      | improvement       |
| V1_original         | summary-matched | 0.20  | 242,610          | 347,291 (leave)   | −104,682 [−116,968, −92,980]  | improvement      | improvement       |
| V1_original         | summary-matched | 0.30  | 292,185          | 468,568 (leave)   | −176,383 [−189,713, −162,843] | improvement      | improvement       |
| V1_original         | summary-matched | 0.50  | 377,466          | 482,448 (contain) | −104,982 [−119,989, −89,476]  | improvement      | improvement       |
| Study4_expanded21ch | contrast        | 0.05  | 57,835           | 97,510 (leave)    | −39,675 [−40,502, −38,711]    | improvement      | improvement       |
| Study4_expanded21ch | contrast        | 0.10  | 80,442           | 161,720 (leave)   | −81,279 [−82,381, −79,901]    | improvement      | improvement       |
| Study4_expanded21ch | contrast        | 0.20  | 125,379          | 290,140 (leave)   | −164,761 [−165,313, −163,659] | improvement      | improvement       |
| Study4_expanded21ch | contrast        | 0.30  | 171,418          | 418,560 (leave)   | −247,142 [−247,969, −245,489] | improvement      | improvement       |
| Study4_expanded21ch | contrast        | 0.50  | 264,231          | 482,448 (contain) | −218,217 [−220,329, −214,727] | improvement      | improvement       |
| Study4_expanded21ch | summary-matched | 0.05  | 138,235          | 165,377 (leave)   | −27,142 [−30,516, −23,013]    | improvement      | improvement       |
| Study4_expanded21ch | summary-matched | 0.10  | 169,578          | 226,015 (leave)   | −56,437 [−63,078, −49,246]    | improvement      | improvement       |
| Study4_expanded21ch | summary-matched | 0.20  | 219,774          | 347,291 (leave)   | −127,517 [−136,851, −117,951] | improvement      | improvement       |
| Study4_expanded21ch | summary-matched | 0.30  | 272,068          | 468,568 (leave)   | −196,500 [−207,596, −184,544] | improvement      | improvement       |
| Study4_expanded21ch | summary-matched | 0.50  | 355,396          | 482,448 (contain) | −127,051 [−139,013, −113,128] | improvement      | improvement       |
| V2_study5           | contrast        | 0.05  | 67,767           | 66,682 (leave)    | 1,085 [−827, 4,357]           | inconclusive     | improvement       |
| V2_study5           | contrast        | 0.10  | 128,699          | 132,515 (leave)   | −3,816 [−6,894, 401]          | inconclusive     | improvement       |
| V2_study5           | contrast        | 0.20  | 242,965          | 264,180 (leave)   | −21,215 [−32,411, −9,003]     | improvement      | improvement       |
| V2_study5           | contrast        | 0.30  | 331,483          | 395,845 (leave)   | −64,362 [−81,171, −47,254]    | improvement      | improvement       |
| V2_study5           | contrast        | 0.50  | 439,377          | 482,448 (contain) | −43,070 [−59,403, −27,504]    | improvement      | improvement       |
| V2_study5           | summary-matched | 0.05  | 66,682           | 66,682 (leave)    | 0 [0, 0]                      | inconclusive     | improvement       |
| V2_study5           | summary-matched | 0.10  | 132,515          | 132,515 (leave)   | 0 [0, 0]                      | inconclusive     | improvement       |
| V2_study5           | summary-matched | 0.20  | 266,863          | 264,180 (leave)   | 2,683 [−551, 7,178]           | inconclusive     | improvement       |
| V2_study5           | summary-matched | 0.30  | 400,582          | 395,845 (leave)   | 4,737 [−6,297, 15,234]        | inconclusive     | improvement       |
| V2_study5           | summary-matched | 0.50  | 481,261          | 482,448 (contain) | −1,187 [−6,308, 4,133]        | inconclusive     | inconclusive      |
| V3_third            | contrast        | 0.05  | 27,887           | 67,562 (leave)    | −39,675 [−40,502, −38,711]    | improvement      | improvement       |
| V3_third            | contrast        | 0.10  | 52,069           | 133,348 (leave)   | −81,279 [−82,381, −79,901]    | improvement      | improvement       |
| V3_third            | contrast        | 0.20  | 102,520          | 264,920 (leave)   | −162,400 [−164,761, −159,331] | improvement      | improvement       |
| V3_third            | contrast        | 0.30  | 149,626          | 396,493 (leave)   | −246,867 [−247,969, −244,663] | improvement      | improvement       |
| V3_third            | contrast        | 0.50  | 247,143          | 482,448 (contain) | −235,304 [−236,091, −233,730] | improvement      | improvement       |
| V3_third            | summary-matched | 0.05  | 47,442           | 70,478 (leave)    | −23,036 [−26,174, −18,963]    | improvement      | improvement       |
| V3_third            | summary-matched | 0.10  | 86,941           | 136,111 (leave)   | −49,170 [−55,902, −41,887]    | improvement      | improvement       |
| V3_third            | summary-matched | 0.20  | 159,252          | 267,376 (leave)   | −108,124 [−119,490, −96,401]  | improvement      | improvement       |
| V3_third            | summary-matched | 0.30  | 218,026          | 398,642 (leave)   | −180,616 [−193,238, −167,364] | improvement      | improvement       |
| V3_third            | summary-matched | 0.50  | 327,382          | 482,448 (contain) | −155,066 [−170,369, −138,163] | improvement      | improvement       |
| V4_fourth           | contrast        | 0.05  | 69,353           | 71,486 (leave)    | −2,133 [−5,648, 2,486]        | inconclusive     | improvement       |
| V4_fourth           | contrast        | 0.10  | 127,386          | 137,066 (leave)   | −9,680 [−17,475, −1,557]      | improvement      | improvement       |
| V4_fourth           | contrast        | 0.20  | 225,250          | 268,225 (leave)   | −42,975 [−57,515, −28,256]    | improvement      | improvement       |
| V4_fourth           | contrast        | 0.30  | 311,081          | 399,384 (leave)   | −88,303 [−103,860, −71,359]   | improvement      | improvement       |
| V4_fourth           | contrast        | 0.50  | 413,286          | 482,448 (contain) | −69,162 [−86,892, −51,584]    | improvement      | improvement       |
| V4_fourth           | summary-matched | 0.05  | 63,377           | 104,705 (leave)   | −41,328 [−41,328, −41,328]    | improvement      | improvement       |
| V4_fourth           | summary-matched | 0.10  | 85,880           | 168,536 (leave)   | −82,656 [−82,656, −82,656]    | improvement      | improvement       |
| V4_fourth           | summary-matched | 0.20  | 130,886          | 296,199 (leave)   | −165,312 [−165,312, −165,312] | improvement      | improvement       |
| V4_fourth           | summary-matched | 0.30  | 175,893          | 423,861 (leave)   | −247,969 [−247,969, −247,969] | improvement      | improvement       |
| V4_fourth           | summary-matched | 0.50  | 265,906          | 482,448 (contain) | −216,542 [−216,542, −216,542] | improvement      | improvement       |

recovers much of the theoretical ranking. **The exceptions are concentrated in Study 5** (5 of 6 cells that do not beat the best constant policy at point estimate; 1 in the original design); the other profiles have a supported advantage over the best constant in every cell except a few contrast cells at the low end of the prior grid (Table 4). Point-estimate counts use strict inequality; Table 4 gives the interval verdicts.

**“Beats always-contain” and “beats best-constant” are different claims.** The complete 50-cell per-prior table uses an explicit, stated criterion (point-estimate comparison, strict inequality — not a statistical significance test, stated plainly so the count is well-defined and auditable), the identical classifier (`LogisticRegression`) and seed-grouped train/test split across all five profiles, and direct class-conditional weighting. At point estimate (strict inequality), the classifier beats always-contain in 50 of 50 cells and the best constant policy in 44 of 50. With cluster-bootstrap intervals over seeds, its advantage over always-contain is a supported improvement in 49 cells and inconclusive at this power in 1; against the best constant it is a supported improvement in 40 cells and inconclusive in 10. No cell shows a supported deterioration against either comparator. The cells without a supported improvement over the best constant are listed in Table 4. These cells lie at the low end of the prior grid or in the Study 5 profile; in one Study 5 cell the best constant policy is always-contain.

Table 4: Cells of Table 3 whose advantage over the best constant policy is not a supported improvement (paired basis, current convention).

| Design      | Regime          | Prior | $R_{\text{clf}}$ | Best constant     | Difference [95% CI]    | vs best constant | vs always-contain |
|-------------|-----------------|-------|------------------|-------------------|------------------------|------------------|-------------------|
| V1_original | contrast        | 0.05  | 99,906           | 97,510 (leave)    | 2,395 [−1,791, 7,564]  | inconclusive     | improvement       |
| V1_original | contrast        | 0.10  | 160,009          | 161,720 (leave)   | −1,711 [−7,827, 5,451] | inconclusive     | improvement       |
| V2_study5   | contrast        | 0.05  | 67,767           | 66,682 (leave)    | 1,085 [−827, 4,357]    | inconclusive     | improvement       |
| V2_study5   | contrast        | 0.10  | 128,699          | 132,515 (leave)   | −3,816 [−6,894, 401]   | inconclusive     | improvement       |
| V2_study5   | summary-matched | 0.05  | 66,682           | 66,682 (leave)    | 0 [0, 0]               | inconclusive     | improvement       |
| V2_study5   | summary-matched | 0.10  | 132,515          | 132,515 (leave)   | 0 [0, 0]               | inconclusive     | improvement       |
| V2_study5   | summary-matched | 0.20  | 266,863          | 264,180 (leave)   | 2,683 [−551, 7,178]    | inconclusive     | improvement       |
| V2_study5   | summary-matched | 0.30  | 400,582          | 395,845 (leave)   | 4,737 [−6,297, 15,234] | inconclusive     | improvement       |
| V2_study5   | summary-matched | 0.50  | 481,261          | 482,448 (contain) | −1,187 [−6,308, 4,133] | inconclusive     | inconclusive      |
| V4_fourth   | contrast        | 0.05  | 69,353           | 71,486 (leave)    | −2,133 [−5,648, 2,486] | inconclusive     | improvement       |

**Sensitivity arm.** Under the alternative fault-cost convention (both fault entries absolute), the classifier beats always-contain at point estimate in 39 of 50 cells, with no supported deterioration against either comparator; the choice of convention is stated in Section 6.

**What this means for the earlier three-profile narrative** (Study 5 evaporates, original contrast reverses to benefit, Study 4/third design un-

affected): the qualitative story survives this recomputation across all five profiles now covered — Study 5 is still the profile with the least cost benefit (and, per the full table, holds the only cell whose advantage over always-contain is inconclusive), and the other profiles have a supported advantage over the best constant in every cell except a few contrast cells at the low end of the prior grid (Table 4); the dollar figures of that narrative are those of Table 3.

**Scope.** Every profile reported in Section 7 has a prior-corrected, positive-fault-loss dollar figure, auditable at the individual-cell level. As stated in Section 6.2, these figures are an illustrative reconstruction, not a calibration.

#### *7.8. A persistence/duration ablation (not a clean causal test of Theorem 2)*

Section 4.1’s Theorem 2 predicts that a *persistent* (state-feedback) disturbance is far more fragile to confound than an equivalent disturbance that does not compound. We attempted to test this directly by holding feature set and nominal footprint parameters fixed and varying only whether the incident’s behavioral shift persists across wakeups or fires once and reverts to baseline. **This comparison does not isolate the mechanism it was designed to test.** A one-off disturbance still changes the order book and can trigger downstream market reactions — a one-off order is not equivalent to “no state feedback,” only to a much shorter and less repeated one. The comparison simultaneously varies repeated forcing, incident duration, cumulative order activity, and the amount of evidence generated, not persistence alone; equal *parameter settings* between the two conditions do not establish equal *realized* footprint (order counts, executed volume, active duration were not measured and matched directly). Lower classifiability from a much shorter, non-repeated incident is an expected finding on general grounds, whether or not state feedback specifically contributes anything.

**What this result actually supports, stated at the correct strength:** shorter, non-repeated incidents were less classifiable than persistent ones in these specific experiments — an exploratory duration/persistence observation, directionally consistent with (though not a clean isolated confirmation of) Theorem 2’s prediction. It does **not**, on its own, establish that state-feedback fragility specifically (as opposed to reduced duration, reduced cumulative activity, or reduced evidence generation generally) explains a specific, quantified share of Section 7’s earlier findings. A design that would isolate the mechanism — comparing reactive against frozen/replayed background market behavior under matched incident order schedules (same commanded

order counts, submitted volume, and active duration in both conditions), with dose (single vs. repeated forcing) varied as a separate factor — is reported in Section 7.10.2, with an inconclusive result.

**Scope of this ablation.** It was run once per condition, before the hardened pipeline (Section 7.10.5), and its robustness check resamples train/test splits of a single dataset rather than drawing independent replicates; its persistent-condition baseline is that earlier run, not the unified basis of the generated classification table. We report it as exploratory.

**Results.** Removing persistence reduced classifiability in both scenarios; in the summary-matched scenario it fell to a level consistent with chance. This is a duration/persistence effect; it is not a clean separable confirmation that state-feedback fragility is doing explanatory work, since duration and dose were not isolated from persistence in this design.

**An additional observation:** the ranking between the two scenarios’ one-off accuracies is technically in the same direction as the original pilot’s ranking, but only marginally — both values sit close to chance rather than showing a clean, large gap. We do not read this as “restoring Theorem 1’s prediction” (since neither scenario was shown to instantiate Theorem 1’s premise in the first place). Removing persistence lowered classifiability in both scenarios, by a larger margin in the summary-matched scenario than in the contrast scenario, consistent with the duration and dose confound identified above; we do not attribute the difference to any isolated mechanism.

#### *7.9. A fourth design: exploratory extension, not a confirmatory test*

Section 7.6 does not claim that the first three injection designs exhaust the space of possible outcomes. To probe this, we built a fourth incident agent (attribute subset {`buy_bias`, `size_mult`, `aggressiveness`}, calibrated against mean spread — a combination distinct from all three prior designs). It produced a **fifth qualitatively distinct outcome**, not a repeat of any existing pattern. This is an exploratory extension, not a confirmatory test: the expectation that a further design might surface a new outcome was general, not a pre-registered, specific prediction, so its being borne out is one more data point in an exploratory process, not confirmatory evidence.

Table 2 gives all five outcomes side by side on the paired basis. The summary-matched regime is the more classifiable one in the original design and in the fourth design (reversed relative to the pilot’s pattern), and the less classifiable one for Study 4, Study 5 and the third design; the fourth design’s reversal is the most extreme of any design. The original and fourth designs

were also regenerated under the verified-clean pipeline using the exact seeds of their earlier runs and produced bit-identical feature arrays, so their earlier data was never contaminated.

The fourth design reverses in the same direction as the original but far more extremely: its confounded case reaches *perfect* classification, the highest of any design tested. **On the paired basis (1000 episodes per class, under the hardened pipeline, with two RNG bugs in this design’s own config file fixed) it has 0 errors in 600 held-out test episodes.** A bootstrap interval on a zero-error test set is degenerate (it can only resample errors that were observed, and none were), so **we report the exact 95% Clopper–Pearson bound given zero observed errors: accuracy at least 0.994 over the 600 test episodes; because the two episodes of a seed are not independent, the conservative bound over the 300 test seeds is 0.988** (Appendix A). This cell is saturated. The mechanism is again directly diagnosable and again a variant of the same underlying pattern: calibrating against one statistic (mean spread) left a large, uncontrolled asymmetry in a different attribute (effective order size, 5.5x vs. 1.5x baseline between the two mechanisms) that leaks strongly through depth-related features, whose pooled-standard-deviation gaps on the calibration data are larger than the calibrated spread gap. In each of the four injection designs, calibration by search left the summary-matched regime classifiable above chance on the paired Era-3 basis (Table 2: the lower end of every summary-matched accuracy interval lies above one half, the lowest being Study 5’s, 0.525), here through a size-based channel rather than the cancellation-rate channel of the third design; for Study 5 the residual signal is weak (0.553 accuracy, interval 0.525–0.580). The four designs are parameterizations of one shared simulation infrastructure (Section 9), so this is a regularity of the constructions tested, not a general law about calibration by search.

**Consequence for Section 7.6’s stopping rule:** this is consistent with, not confirmatory proof of, that justification. The instability observation rests on five outcomes across four designs sharing one infrastructure, with no sign of convergence toward a single stable pattern; the complete characterization of that instability remains open (Section 7.6), and we do not treat this fourth design as having tested or confirmed anything beyond what continued exploration would be expected to find.

### 7.10. Negative control, cost-pipeline checks and robustness audits

This subsection reports eight further checks (Sections 7.10.1–7.10.8), with genuinely mixed results — clean validations alongside a materially worse finding and one still-open complication — followed by the final end-to-end regeneration under the hardened pipeline (Section 7.10.9).

**7.10.1 Negative control — no detectable accuracy signal; AUC inconclusive.** Two incident-agent variants with *identical* observable order-generation kernels (same distribution over rate, size, side, aggressiveness, and cancel behavior; differing only in an internal, behaviorally-inert label) were run through the full classification pipeline. **K/F seed assignment is interleaved from one shared pool:** drawing K and F episodes from two fixed, disjoint random-seed blocks produces a genuine, if small, systematic effect (a controlled test with real vs. randomly-shuffled labels on the same episodes; with bit-identical exogenous-stream coupling, mean accuracy over repetitions falls noticeably below chance), with no orientation or sign-convention bug in the classifier or label assignment.

Repeated cross-validation on one fixed dataset is the wrong tool here: it measures variance across partitions of the *same* episodes, not uncertainty over new, independent episodes — folds and repeats share data, so its interval (repeated 5-fold on a large-N, interleaved-seed dataset, very narrow for both accuracy and AUC) understates true uncertainty. **The evaluation instead uses 20 genuinely independent replicate datasets** (fresh episodes each time — 100 episodes per replicate; confirmed by direct source inspection that each replicate’s K and F halves are drawn from disjoint seed pools with no shared exogenous stream, so no pairing structure exists that could be broken by the standard unpaired train/test split used here), with a between-replicate confidence interval computed across those 20 replicate-level values rather than within one of them.

**Interval bounds, computed two ways ([L7-18]).** The between-replicate 95% intervals, by normal approximation and by percentile bootstrap over the 20 replicate values, agree closely. **Checked against the pre-specified equivalence margin  $[0.45, 0.55]$ : accuracy’s interval is fully contained within the margin; AUC’s interval is not** — its lower bound falls just outside 0.45. AUC’s point estimate is inside the margin but its interval is not. The valid between-replicate interval is much wider than the within-dataset CV interval. A further distinction worth being precise about: this 95%-interval-containment check is a conservative basis for equivalence, distinct from a formal two-one-sided-tests (TOST) equivalence

test at  $\alpha = 0.05$ , which would use a 90% interval instead — an independent check using a one-sample  $t$ -procedure on the reported summary statistics gives the same reading under either framing (accuracy interval inside the margin, AUC interval not), but the two are not interchangeable in general and we do not claim to have run a formal TOST here. **For accuracy, the between-replicate confidence interval lies inside the pre-specified equivalence margin, so the tested procedure shows no detectable accuracy signal at this margin and sample size; for AUC, the interval extends below the margin, so equivalence to chance is not established for AUC, and the small shortfall of its mean below one half is not diagnosed. We did not run a formal two-one-sided-tests procedure.** This finding concerns the mean performance of this specific fitted procedure at this sample size, not a claim about equality of the full observation laws or a bound applicable to every possible classifier.

**7.10.2 Dose/persistence isolation — inconclusive, reported as such.** The redesigned reactive-vs-replayed comparison (Section 7.8’s follow-up, isolating state feedback from duration/dose) was built and run, catching and fixing two further implementation bugs (an RNG-desynchronization issue and a latency-double-counting issue) along the way. **The 4-cell result at  $N = 30$  per cell is genuinely inconclusive** — we do not have a large enough sample to distinguish a real feedback effect from noise in this design, and we report this as an open question rather than force a conclusion. Section 7.8’s duration/dose-confound correction stands; this experiment does not yet resolve it either way.

**7.10.3 Frozen confirmation set — not provenance-clean.** Section 7.10.5 found that this set was generated under an internally mixed pre-/post-patch provenance, so it is not used as evidence; a verified-clean regeneration was produced under the hardened pipeline (Section 7.10.5).

**7.10.4 Cost pipeline — prior-corrected threshold and positive fault loss.** Section 7.7’s table applies two separate corrections together: the prior-shift threshold correction (Section 6) and the positive fault loss (Section 6.3’s audit). The decomposition identity for the positive-fault-loss model used in Table 4 is:

$$R_{\text{oracle}} = p C_K + (1 - p) H_F,$$

$$R(\pi) - R_{\text{oracle}} = p (H_K - C_K) \cdot \text{FNR} + (1 - p) (C_F - H_F) \cdot \text{FPR},$$

with cost-optimal threshold  $q^* = \frac{C_F - H_F}{(H_K - C_K) + (C_F - H_F)}$  — the zero- $H_F$  formulas in Section 6 remain correct *for that section’s own toy scenario*, where  $H_F = 0$  by construction, but should not be read as governing Section 7’s positive-fault-loss model. With both corrections: Study 5 holds most of the cells without a supported improvement over the best constant policy, and 1 of its cells has an advantage over always-contain that is inconclusive at this power; the other profiles have a supported advantage over the best constant in every cell except a few contrast cells at the low end of the prior grid (Table 4). Study 4 and the third design’s *qualitative* advantage survived the threshold correction, though — per a full per-episode check — only one specific cell (Study 4’s contrast regime) was confirmed literally unchanged at the individual-decision level; the other cells in these two profiles shifted modestly in magnitude while keeping the same sign.

**A further nuance, not to be smoothed over: the prior-shift fix is not a pure improvement — it trades one error type for another.** Reporting false-negative alongside false-positive rates shows that the fix which lowers false-positive rates across every profile *raises* false-negative rates substantially — for example, in Study 5’s contrast regime the false-positive rate falls sharply while the false-negative rate rises sharply. This makes sense mechanically (a threshold correctly adjusted for a low deployment prior becomes much more conservative about calling “compromise,” which necessarily lets more true compromises through uncaught) but means the corrected pipeline should not be described as simply “more accurate” — it is more accurate about *cost*, given the specific cost triple used, but the error profile has shifted, not uniformly shrunk, and a deployment with a different cost triple (or a genuine preference for erring toward containment) would need to re-derive its own threshold rather than assume the correction is cost-triple-independent.

**A separate methodological subtlety surfaced while re-deriving the FPR/FNR tables:** the cost-decomposition identity  $R(\pi) - R_{\text{oracle}} = p(\text{HARM} - C_{OP}) \cdot \text{FNR} + (1 - p)\text{DISRUPT} \cdot \text{FPR}$  holds exactly (to floating-point precision) when checked against the *realized* resampled prior in a given table row, but checking it against the *nominal* target prior  $p_k$  instead produces spurious apparent violations — a reminder that any reader attempting to independently verify this paper’s cost tables should reconstruct the realized class balance from the raw predictions, not assume the nominal prior was achieved exactly in every finite resample.

**7.10.5 RNG and provenance audit.** Beyond the originally-disclosed

`get_wake_time()` issue, further unseeded-RNG bugs were found and patched. Bit-identical cross-process reproducibility was demonstrated only after all of these fixes were applied. **A subsequent investigation then found that our own patch-verification process was itself unreliable:** worker processes performed their own lazy, per-worker import of the affected modules and cached that import for their entire lifetime, so results generated *after* the patch was written to disk could still reflect pre-patch code if the worker that produced them had already cached the old version — a subtler failure mode than simple parent-process staleness, confirmed by direct reproduction and fixed with a forced fresh re-import plus a behavioral fingerprint check (`verify_patch_active()`) that verifies all the RNG patches are actually active in a given worker, independent of file timestamps. **This fingerprint-based fix is not sufficient on its own:** reimporting a module does not update references already held by objects or workers created before the reimport, and a further investigation confirmed this concern was valid and worse than described — the fingerprint check itself gave a false positive in one deliberately-constructed test (following a real mid-run file reversion), and a second failure mode was found where a re-exporting parent package silently defeated what looked like a “fresh” submodule import. **Both are now closed** by killing and respawning every worker process fresh per task (`maxtasksperchild=1`), verified to eliminate the gap completely *for the specific caching failure mode diagnosed*. **A further precision: `maxtasksperchild` controls worker recycling, not the multiprocessing start method** (`fork` vs. `spawn`) — a forked worker can still inherit a stale parent’s already-imported state even when recycled per-task, whereas `spawn` creates a genuinely fresh interpreter. **A project-wide audit of every production script’s actual start method** finds three eras: early scripts use plain `fork` with no hardening at all; a middle group uses `spawn` but without `maxtasksperchild`; only scripts written from the point this fingerprint fix was introduced onward use the fully-hardened combination (`spawn` + `maxtasksperchild=1`). **Provenance of each result.** The ten headline classification cells (Table 2) and the cost table built on them were generated with the hardened pipeline (`spawn`, one task per worker, runtime patch fingerprint passing for every task), on paired seeds. The measured fault-side harm  $H_F$  used by the cost table is computed per fault episode from those same episodes, whose trajectories were recovered and checked bit-identical to the stored features, not from an earlier run. Every other ABIDES result in Section 7 carries its generation era in the provenance register (Appendix A);

results from the two earlier eras (Sections 7.1–7.8 and the severity and dose experiments) were produced before the RNG patches or with scripts that would not detect a mid-run patch, and are reported as exploratory. The frozen confirmation set of Section 7.10.3 is not used.

**7.10.6 Signed-volume sign convention — root cause and fix.** The “signed executed volume” feature used in Study 4 (Section 7.2) is computed as `buy_vol - sell_vol` over executed trades in a time bin, where `buy_vol/sell_vol` are summed by the `side` field ABIDES’s order book logs for each execution. **The inversion’s exact mechanism, traced to source:** ABIDES’s order book logs the *passive/resting* order’s side at execution, not the aggressor’s — `side="SELL"` if `order.side.is_bid()` else `"BUY"`, where `order` is the incoming aggressor order, so an aggressor buy (hitting a resting ask) is logged as `side="SELL"` (the resting order’s side), and vice versa. The feature’s code therefore computes (`aggressor-sell-initiated volume`) — (`aggressor-buy-initiated volume`), the exact negation of what the variable name and this paper’s documentation describe as “buy-initiated minus sell-initiated” pressure. **The fix is a single negation — signed\_exec\_vol\_CORRECT** = sell-labeled qty – buy-labeled qty in the current logging convention, equivalently, flip the sign of every value the current code produces. This does not require access to any information a real market participant couldn’t have: the passive/aggressor distinction is recoverable from ordinary trade-and-quote data (compare execution price to the prevailing best bid/ask at execution time), so the feature remains operationally valid once correctly signed — this was a pure implementation bug, not a hidden-information problem. This sign error has zero effect on any reported accuracy or AUC (a linear classifier learns the data’s true orientation regardless of which label the documentation assigns it), so no classification result in this paper depends on it.

**7.10.7 Severity extrapolation — magnitude generalization within the same mechanisms, not held-out mechanisms.** The table below varies incident *magnitude* outside a training range within the *same* four constructions (the original design, Study 4, Study 5 and the third design, all parameterizations of one shared infrastructure), not generalization to a mechanism never seen during training or calibration. A genuine held-out-*mechanism* test (training on three designs, evaluating on a fourth never seen during training) is reported separately below in this same subsection. The fourth design (Section 7.9) has no row in the severity-extrapolation table; it is covered only by the leave-one-design-out test, so the two tests cover

different sets of four.

| Design / feature set                | In-range (held-out) | Out-of-range (extrapolated magnitude) | Pattern                                                                                |
|-------------------------------------|---------------------|---------------------------------------|----------------------------------------------------------------------------------------|
| Original, 4-channel                 | [L7-21]             | [L7-21]                               | Improves (expected: higher severity is an easier signal, not evidence of memorization) |
| Original, Study 4 expanded features | [L7-21]             | [L7-21]                               | Saturated                                                                              |
| Study 5                             | [L7-21]             | [L7-21]                               | Accuracy drops substantially out-of-range                                              |
| Third design                        | [L7-21]             | [L7-21]                               | Negligible drop                                                                        |

Three of four rows generalize cleanly or improve out-of-range, the expected pattern when the held-out range is more severe, not evidence of memorization. **Study 5 moves in the opposite direction, and we state two candidate explanations without claiming to have established which is correct:** (a) Study 5’s in-range estimate, on a very small test set, was an optimistic small-sample draw, and the larger out-of-range test is closer to Study 5’s weak discrimination on the paired basis (Section 7.2); or (b) this reflects a genuine distribution shift specific to this design. We do not have matched sample sizes between the in-range and out-of-range tests, which would be needed to distinguish these two explanations; both remain open pending a dedicated follow-up with matched sample sizes.

**Leave-one-design-out generalization, distinct from the severity-extrapolation table above (compromise and fault remain the two known cause labels throughout):** training on three of the four designs and evaluating on the fourth, held out entirely from training and calibration. **A below-one-half AUC is not “no discriminatory structure”** — for a real-valued score,  $\text{AUC}(-s) = 1 - \text{AUC}(s)$ , so an AUC well below one half corresponds to a *reversed* ranking that is strongly discriminative under the opposite orientation; when its interval lies entirely below one half, chance

does not explain it. Such cells are compatible with the held-out design’s score orientation failing to transfer from the three training designs, rather than the classifier learning nothing transferable. (To be explicit about what we are not claiming: reversing a score after inspecting held-out labels would not be a valid way to report a deployment improvement — the complement is a diagnostic interpretation of what happened, not a proposed correction to apply going forward.) Per-cell accuracy, AUC, bootstrap intervals over the held-out test set and train/test counts for all eight held-out cells: [L7-22]. **We read the test as evidence that classifier score orientation is substantially construction-specific and does not reliably transfer across separately designed mechanisms — sometimes failing to transfer at all, and in the third design’s case transferring in reversed form — which is a more specific finding than “no transferable signal,” and still a genuine limit on generalizing any single construction’s classifier to a different one.**

**The root cause of the third design’s reversal.** The pipeline itself is clean: the classifier (`LogisticRegression`), feature set (the identical original-channel feature set — summaries of spread, depth, cancellation rate and mid-price — across all four designs, confirmed by matching array shapes), and train/test split are confirmed identical across all four held-out configurations, with the held-out design’s data loaded from a separate file never passed to `.fit()` — no leakage or pipeline inconsistency. The reversal is **not** a signed-volume sign-convention issue, since signed volume is not among the features used in this specific experiment. Examining the pooled classifier’s learned coefficients (trained on the other three designs) against the third design’s own class-conditional feature means: only some of the features show a  $K-F$  direction in the third design opposite to what the pooled classifier learned to associate with  $K$  — not a majority of features, so the reversal is not simply “every feature flipped,” but concentrated in a smaller number of high-coefficient-magnitude features, particularly `spread_mean` (the pooled classifier’s largest-magnitude coefficient). **The mechanistic explanation:** the third design’s incident agent has no directional (`buy_bias`) dimension at all — side is an unbiased coin flip for both  $K$  and  $F$  — so its actual  $K$ -vs- $F$  distinction is driven entirely by size, rate, and cancellation-rate differences, a structurally different axis from the other three designs’ predominantly directional distinction. A classifier trained mostly on directional signal, applied to a design whose real distinguishing signal lies elsewhere, does not merely fail to generalize — it applies a decision rule that happens to be anti-correlated

with the third design’s actual (non-directional) mechanism, producing the observed reversed ranking rather than a null result.

**7.10.8 Error rates alongside cost — the decomposition holds exactly, under the correct (positive-fault-loss) identity.** Reporting false-positive and false-negative rates alongside accuracy/AUC for every cost table confirms the identity  $R(\pi) - R_{\text{oracle}} = p(H_K - C_K) \cdot \text{FNR} + (1 - p)(C_F - H_F) \cdot \text{FPR}$  exactly (verified to within floating-point precision; the zero-fault-loss form of Section 6 applies only to that section’s toy scenario) and shows that threshold sensitivity — how much a policy’s cost depends on exactly where the decision threshold is set — is concentrated almost entirely in the weaker classifiers (Study 5 and the original contrast construction). **Study 4 and the third design’s separated-regime error rates are near zero, but their *confounded*-regime FNRs are not** (Table 3 lists the per-cell verdicts; the false-negative rates of these cells are nonzero at every prior and largest at the lowest priors).

**Two different thresholds appear in this paper and should not be conflated.**  $q^* = (C_F - H_F) / [(H_K - C_K) + (C_F - H_F)]$  (Section 6) is fixed *on the deployment posterior* — it depends only on the loss table, not on the prior  $p$ , and is the threshold applied to a correctly prior-adjusted score. A separate quantity,  $s^*(p)$ , is the *equivalent* cutoff applied directly to a raw score calibrated to a different (e.g., balanced 50/50) training prior — this one *does* vary with  $p$ , since it must absorb the prior-shift correction that  $q^*$  does not need to. Where this section and Section 6 refer to “the cost-optimal threshold” without further qualification, this means  $q^*$  applied to the prior-adjusted score, not  $s^*(p)$  applied to the raw one — the two coincide only when  $p$  equals the training prior exactly.

**A further caution:** mean FPR and mean FNR across five priors, multiplied by a mean prior, do not reproduce mean cost — the cost-optimal threshold itself changes with the prior, so per-prior rates and per-prior weights are associated, not independent; the per-prior decomposition (not shown in full here) should be checked before aggregating, and the by-design-and-regime means reported in Section 7.7 are simple averages of the per-prior figures, not a re-derivation from averaged rates.

**7.10.9 Final end-to-end regeneration under the hardened pipeline.** The run-manifest audit (Appendix: Reproducibility) established that the earlier cost table’s *classifier training data* predated the RNG-audit patches for every profile, and had never been regenerated under the fully-hardened (spawn + maxtasksperchild=1 + runtime fingerprint verification) pipeline

for Study 5, the third design, or Study 4. **All three were regenerated end-to-end at large  $N$  under the hardened pipeline, before the paired basis of Table 2 was built.** The table below flags three of the six regenerated cells as changed.

| Design / regime             | Pre-patch<br>acc/AUC | Hardened, large<br>$N$ | Change                                |
|-----------------------------|----------------------|------------------------|---------------------------------------|
| Study 5,<br>separated       | [L7-24]              | [L7-24]                | Changed (sample<br>size also differs) |
| Study 5,<br>confounded      | [L7-24]              | [L7-24]                | Changed (AUC)                         |
| Third design,<br>separated  | [L7-24]              | [L7-24]                | No meaningful<br>change               |
| Third design,<br>confounded | [L7-24]              | [L7-24]                | <b>Lower</b>                          |
| Study 4,<br>separated       | [L7-24]              | [L7-24]                | No meaningful<br>change               |
| Study 4,<br>confounded      | [L7-24]              | [L7-24]                | No meaningful<br>change               |

**Two findings.** First, **the third design’s near-perfect separated-regime result is confirmed genuine** — it survived the patch essentially unchanged, ruling out the RNG bug as its explanation, which is exactly the kind of targeted test that distinguishes a real result from an artifact. Second, **the same design’s confounded-regime result did not survive as well:** its apparent near-separability was partly inflated by the unpatched RNG, and the corrected figure is meaningfully lower. Study 5’s regenerated cells read higher than its pre-patch ones, but its pre-patch run was smaller than the regeneration, so that change is not attributed to hardening; Study 5 remains the weakest of the five profiles. Propagating the three changed cells through the cost pipeline shifted  $\hat{R}_p$  in both directions, most at the third design’s confounded regime at the highest prior ([L7-24]); the cost table of Section 7.7 is built on the paired basis, not on these regenerated cells. One production code change was made in the course of this work: the third design’s configuration builder, which the manifest audit found had never been patched for the RNG bugs at any point in the project, was permanently patched (Appendix).

### 7.11. A question left to Paper 2

The cells of Section 7.7’s cost table in which the fitted classifier’s advantage over the best constant policy is not a supported improvement lie at the low end of the prior grid or in the Study 5 profile (Table 4): passive observation adds no supported value there.

Two results of this paper bound what it can contribute to that question. **First**, whether  $K$  and  $F$  are observationally confounded (Theorem 1’s condition,  $\theta_K C b_K = \theta_F C b_F$ ) is a property of the fixed mechanism instances and the observation law and does not depend on the prior  $p$ ; what depends on  $p$  is the value of resolving the ambiguity, Proposition 1’s optimal ambiguity cost  $\min\{(1-p)\text{DISRUPT}, p(\text{HARM} - C_{OP})\}$ , which is zero exactly when  $p \in \{0, 1\}$ . **Second**, Section 4.1’s Theorem 2 shows that even within the linear model exact confounding is not generic under state-level injection, and none of the ABIDES constructions of this section was shown to instantiate Theorem 1’s premise (Section 7.3). This paper therefore defines no confounded region for a real incident and evaluates no verification channel: neither a verification channel’s accuracy, cost, or timing nor a value-of-information comparison against the passive alternative appears anywhere in it.

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## 8. Discussion

The identifiability boundary from the linear model is not worthless — Sections 3–6 show it is exact and reproducible within its stated scope, and it remains a legitimate diagnostic wherever that scope holds (plausibly, small-perturbation regimes around a steady-state book). **But Section 4.1 already shows that scope is narrower than Theorem 1 alone suggests:** confounding that survives a one-shot observation check is not confounding that survives feedback into realistic dynamics, and the gap between the two is governed by a well-understood classical quantity (the unobservable subspace of  $(A, C)$ ) rather than anything exotic. This gives a first, purely linear-model reason to distrust a naive reading of Theorem 1 as operationally robust, before nonlinearity is even introduced. Section 7’s empirical study of ABIDES adds a second, independent caution, at the correct strength: the study was never a test of Theorem 1, since none of the tested constructions were shown

to instantiate the theorem’s actual premise (full trajectory-distribution equality) — what it shows is that attempting to construct such an instantiation in a realistic multi-agent market is itself difficult and sensitive to construction choices that are, per a direct audit (Section 7.3), mostly representable-but-unexplored on the linear-model side rather than genuinely structurally absent from it. **Calibration-by-search is not a general structural dividing line between linear and nonlinear systems: this specific, known-parameter linear construction has an analytic matching solution, while this specific ABIDES construction required numerical search, a fact about these two particular constructions, not a general property of linearity versus nonlinearity. A third caution, methodological rather than theoretical (Section 6): a decision-cost claim is only as trustworthy as the policy used to generate it,** and a plausible-looking, sufficiently surprising empirical result should prompt checking the decision rule before checking the theory. The broader methodological point across all three cautions, applicable beyond this paper: **an identifiability boundary proved for a linearized model is a claim about that model, not a claim about deployment, until the transfer is actually established,** and neither the theoretical fragility (Section 4.1) nor the empirical construction-dependence (Section 7) nor a cost-analysis finding should be taken on the strength of a single striking result without independent verification.

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## 9. Limitations

The paper has a proved theoretical core, a pilot-scale simulation consistent with it inside the linear model, and an exploratory, non-confirmatory study of observation matching in a nonlinear market simulator, which is not a validation of the linear model. Its main limitations are:

1. **Scenario coverage.** The incident-generation layer is wired into ABIDES, a real agent-based market simulator, with realistic order-book mechanics and a background population of noise, value and market-maker agents (Section 7), but only two mechanisms (compromise, fault) are treated; the full scenario-family taxonomy (behavior, regime, and combinations) is not covered.

2. **No linearization bridge.** No linearization bridge (operating point, Jacobian, or approximation-error bound connecting the linear model to ABIDES’s dynamics) has been derived: Section 7 is an independent empirical comparison, not a validation of the linear model by the nonlinear one. What Section 7 does provide, and what should not be confused with linearization validation, is a five-part sensitivity analysis of how ABIDES classification/cost behavior responds to feature, footprint, and classifier choices, with bootstrap or between-replicate confidence intervals where they are reported (the paired-basis classification and cost tables, which carry Studies 4 and 5, the negative control of Section 7.10.1 and the leave-one-design-out test of Section 7.10.7); Study 1, Study 2, Study 3, the footprint sweep of Section 7.5, the persistence ablation of Section 7.8 and the isolation redesign of Section 7.10.2 are reported without intervals. A linearization claim would require actually deriving that bridge.
3. **Realistic cost calibration.** The sensitivity sweep (Section 6.1) shows the qualitative *linear-model* finding is not an artifact of the specific cost triple, but real market-impact calibration is still needed, including for the ABIDES-derived costs given Section 7’s reversal.
4. **Baseline breadth.** The chance-level result under linear-model confounding is confirmed across four classifier families (Section 5.1). On the ABIDES side, Study 2 (Section 7.2) tested four classifier families (logistic regression, Gaussian naive Bayes, random forest, and a threshold rule) on the original construction only; the same sweep has not been run on the other three ABIDES designs (Study 5, third design, fourth design), so classifier-choice sensitivity is confirmed for one construction, not across all four.
5. **Independence of implementations.** A separately coded linear simulator exists (Appendix A) but its results are not reported here. For ABIDES, an independence audit of every retained result found that all four injection designs (original, Study 5, third, fourth) share the same underlying `config_builder`/kernel/feature/cost-pipeline code — they are independently-*parameterized* variants, not independently-*coded* implementations. Zero results in this paper’s ABIDES component qualify as a genuine second implementation in the sense of a separately coded implementation; a bug shared across all four designs’ common infrastructure would not be caught by any comparison currently in this paper. A true second implementation — a separately-coded ABIDES harness,

not merely a different calibration target within the existing one — is a real gap.

6. **Feature and population sensitivity.** The ABIDES result depends on a fixed, hand-chosen feature set and a single population configuration; the companion validation report (not reproduced in this manuscript) notes that mid-price had to be added to the feature set after its absence made the separated case undetectable by construction.
7. **Disentangling Theorem 2’s state-feedback fragility from Section 7’s empirical factors.** An exploratory persistence ablation (Section 7.8) found an effect directionally consistent with Theorem 2’s prediction (removing persistence lowered summary-matched classifiability to a level consistent with chance in one exploratory, pre-hardening run), but this design conflates persistence with incident duration and cumulative activity, so it does not isolate the mechanism cleanly. A reactive-vs-replayed redesign (Section 7.10.2) built to isolate it is inconclusive at the sample size tested ( $N = 30/\text{cell}$ ); this remains open.
8. **An open-ended design space.** A fourth injection design (Section 7.9) found a fifth distinct outcome, so the space of possible outcomes remains open-ended rather than closed by any fixed number of designs; a systematic characterization (a structured sweep over attribute-subset/calibration-target combinations, Section 7.6) is future work.
9. **Provenance of Sections 7.1–7.8.** Nearly all ABIDES results in Sections 7.1–7.8 predate the RNG fix (Section 7.10.5) and verified-clean generation, and should be read with the caveat stated at the opening of Section 7. The frozen confirmation set (Section 7.10.3) is not provenance-clean and is not relied on.

None of these limitations is small. Together they bound what the paper establishes: a sound, provable theoretical core (Sections 3–6) whose practical relevance is genuinely uncertain and possibly narrower than assumed (Section 7).

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## 10. Conclusion

We gave a linear-Gaussian model of gateway-observable order flow and proved an exact fixed-instance equality criterion —  $\theta_K C b_K = \theta_F C b_F$ , together with a separate rank-based characterization of which severity/direction

pairs *can be made* to satisfy it — for when a cyber compromise and a benign fault are observationally indistinguishable from an operator’s passively available evidence, together with a proof that this imposes an unavoidable cost floor — equal to the best-achievable-using-observations risk minus the oracle’s risk — on any policy relying on that observation alone. This part of the result is exact. The optimal ambiguity cost is, under an illustrative (not empirically calibrated) Almgren–Chriss-based dollar reconstruction, approximately [L7-3c] per incident-episode at the illustrative scenario’s assumed base rate. We then showed, without leaving the linear model, that this confounding is itself far more fragile than Theorem 1 alone suggests: modeling the incident as a state-level rather than output-level disturbance requires confounding to lie in the classical (Kalman) unobservable subspace of the dynamics — generically the trivial subspace, though the specific pair-relevant subspace, not the entire kernel of  $C$ , is what actually matters (Section 4.1). We then conducted a controlled empirical study — explicitly not a test of Theorem 1, whose premise (full observation-trajectory-law equality) none of our simulated constructions were confirmed to satisfy — of how incident-construction, feature-access, and calibration choices affect classification and decision cost in ABIDES, a realistic agent-based market simulator. Across four injection designs (parameterizations of one shared infrastructure), several qualitatively distinct classification profiles emerged, sensitive to choices that a direct audit (Section 7.3) found to be mostly representable-but-unexplored on the linear-model side rather than genuinely absent from it. The need for calibration by search is a fact about the two constructions used, not a general dividing line between linear and nonlinear systems (Section 8). We conclude that the linear model’s identifiability and cost-floor results are exact and reproducible within the regime they are proved for; that this regime is narrower than a first reading suggests, both because of Theorem 2’s state-feedback fragility and because engineering a valid nonlinear analog of exact confounding turns out to be a genuinely difficult, easy-to-get-wrong task in the specific constructions attempted here, not a demonstrated general property; and that a decision-cost claim built on a misspecified policy can manufacture findings in either direction — sometimes inflating harm from nothing, sometimes hiding real benefit (Section 6). The paper’s contribution is a theoretical core plus a controlled, honestly-scoped empirical study of observation-matching and decision cost in a simulated market — not a demonstrated failure of linear theory under nonlinear conditions, and not a general claim about calibration difficulty beyond the specific constructions

tested. The cells in which the fitted classifier adds no supported value over the best constant policy (Section 7.7) leave a question to the companion paper (Paper 2), stated in Section 7.11 together with what this paper does not claim about it.

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## **Data and code availability**

The simulation code, the experiment configurations, the provenance register and the result files from which every number in this article is generated are held in the author’s project repository. They will be deposited in a public archive on acceptance, with a persistent identifier (DOI to be determined). Until then they are available from the corresponding author on reasonable request. The ABIDES market simulator is open-source software; the patches applied to it are part of the deposit.

## **CRedit authorship contribution statement**

**Chiheb Mejrissi:** Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

## **Declaration of competing interest**

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used large-language-model assistants (Claude, Anthropic) to draft and revise parts of the text and to write and review parts of the analysis code. After using these tools, the author reviewed, checked and edited the content as needed and takes full responsibility for the content of the published article.

## Appendix A. Reproducibility

**A concurrency defect in ABIDES log writing.** ABIDES’s `Kernel.write_summary_log` names its output directory by wall-clock timestamp at one-second resolution; parallel workers finishing in the same second collide on `./log/<timestamp>/` (a time-of-check/time-of-use race). The failure is loud — the affected episode crashes — and never corrupts data silently, so no retained result is affected. The fix, applied to all runs on the unified basis, gives each worker its own working directory.

**Test denominators of the saturated cells.** Every cell of Table 2 has 1000 episodes per class and 600 held-out test episodes from 300 test seeds (seed-grouped split). 1 cell has zero test errors: the fourth design’s summary-matched regime. Its exact 95% Clopper–Pearson lower bounds are given over the test episodes and, because the two episodes of a seed are not independent, conservatively over the test seeds (Table A.5). The two near-perfect contrast cells have 1 (Study 4) and 1 (third design) test errors; their exact Clopper–Pearson intervals over the test episodes are 0.991–1.000 and 0.991–1.000.

Table A.5: Exact Clopper–Pearson lower bounds for the zero-error cell (paired basis).

| Design    | Regime          | Test episodes | Exact lower bound | Test seeds | Exact lower bound |
|-----------|-----------------|---------------|-------------------|------------|-------------------|
| V4_fourth | summary-matched | 600           | 0.994             | 300        | 0.988             |

Sections 5–6 results were generated by two self-contained Python scripts (`sim.py`, `cost_quantification.py`, plus `cost_sensitivity.py` and `baseline_robustness.py` for the robustness checks in Sections 5.1/6.1, each subsequently re-run across five seeds; those intervals are not reported in this paper), using NumPy and scikit-learn, fixed random seeds. **Section 4.1’s**

**Theorem 2** was validated by two further scripts (`state_vs_output_injection.py` and `coupled_A_test.py`), the first of which surfaced and explained a null result (the paper’s own  $A = 0.6I$  trivially preserves confounding under feedback) before the second, using a genuinely coupled  $A$ , confirmed the predicted fragility — both scripts and their output are included for transparency about that two-step process rather than presenting only the confirming result. Section 7’s ABIDES results were generated by an incident-injection layer built on a local clone of ABIDES (`abides-core`, `abides-markets`), full harness (`incident_agent.py`, `config_builder.py`, `features.py`, `regimes.py`, `calibrate.py`, `run_episode.py`, `batch_run.py`, `task3_classification.py`, `task4_cost_quantification.py`), a separately coded linear-model simulator (`independent_linear_sim.py`; results not reported in this paper), three further ABIDES injection designs beyond the original, all parameterizations of the same shared infrastructure (Study 5, the third design of Section 7.5, and the fourth design of Section 7.9 — four injection designs in total including the original), a footprint sweep, a per-feature ablation, and an Almgren–Chriss illustrative dollar-scale reconstruction (Section 6.2; not a calibration), all detailed in the companion validation report. **Supplementary data:** `paired_basis_cost_table.csv` (all 50 design  $\times$  regime  $\times$  prior cells of Table 3, with classifier, constant and oracle costs, FPR/FNR, threshold, test counts, cluster-bootstrap intervals, interval verdicts and both point-estimate booleans with the stated criterion), its rebaselined counterpart, `paired_basis_classification.csv` and `h_f_era3.csv`. The leave-one-design-out cells of Section 7.10.7 are added once promoted.

**A run-manifest audit maps each published result to the run that produced it; it surfaced a provenance gap, closed by the regeneration of Section 7.10.9.** This project’s ABIDES fork is based on commit `f9cbe51342b7dedd9587e4e069040d68a5c6477f` (2023-12-13), with four locally-modified files (three implementing the RNG-audit patches A/B/C, one an unrelated fill-price logging fix). **The classifier training feature data originally underlying the earlier cost table — every one of the five profiles, both regimes — was generated between 2026-09-08 and 2026-09-09**, before the RNG-audit patch existed at all, and in three of five cases (Study 5, the third design, the fourth design) before that specific design’s own config-builder file was individually patched, with the third design’s config-builder unpatched before the regeneration of Section 7.10.9. The cost table of Section 7.7 no

longer rests on that data: it is built on the paired basis, whose classification cells and measured  $H_F$  come from the same hardened-pipeline episodes (Section 7.10.5).

**All five profiles are regenerated end-to-end at large  $N$  under the fully-hardened pipeline (Section 7.10.9), and `config_builder_v3.py` is permanently patched; the paired basis of Table 2 was then generated under the same pipeline.** Three of six regenerated cells are flagged as changed (Section 7.10.9): the third design’s confounded regime fell, revealing that part of its apparent near-separability had been an RNG artifact, and Study 5’s cells read higher, a change that also reflects a larger sample and is not attributed to hardening. Section 7.10.9 reports the full before/after comparison and the resulting cost-pipeline shifts. V1 and V4’s large- $N$  reruns reproducing their originals closely would *not* have predicted the third design’s confounded-regime change, so the gap could not be assumed clean by extension from those two designs.

Two reproducibility notes (the ABIDES fork is based on `jpmorganchase/abides-jpmc-public` commit `f9cbe513...` dated 2023-12-13, with the four locally-modified files listed above — three RNG patches and the fill-price logging fix — and the `str_to_ns` fix described below): (1) we found and fixed a real bug in ABIDES’s `str_to_ns()` utility under pandas  $\geq 2.x$  that silently mis-scaled every simulated time interval by 1000x with no crash or warning; any prior result produced with this codebase under a modern pandas version should be treated as suspect until checked against this fix. (2) We separately found a bug in ABIDES’s `get_wake_time()` utility that draws from an unseeded global random number generator, meaning identical seeds do not produce bit-identical episodes across separate process invocations (only within one) — this bug, along with further unseeded-RNG bugs (Section 7.10.5), is patched, and bit-identical cross-process reproducibility was demonstrated after all of these fixes were applied. **A subtler defect concerns the verification process itself:** forked worker processes cache their first lazy import of the patched modules for their entire lifetime, so a result generated after the on-disk patch could still reflect pre-patch code depending on exactly when the worker that produced it first imported the module — this affected the frozen confirmation set, which is not uniformly post-fix and is not used; a hardened pipeline (spawn, one task per worker, forced fresh import, and a behavioral fingerprint check independent of file timestamps) was built, and the headline classification cells and the cost table are generated with it (Section 7.10.5). The other ABIDES results of Sections 7.1–7.8 predate it: they

are statistically valid but not bit-for-bit reproducible from the stated seeds, and are reported as exploratory.

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